ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

Department of Physics and Astronomy "A. Righi"

Theoretical Physics

Statistical Field Theory

Lectures held by
Prof. Francesco Ravanini

Notes by
Gioele Mancino

Academic Year 2025/2026

Contents

Introduction

1 Review

In this chapter we will review some basic concepts of statistical mechanics, which will be useful for the rest of the course. We will start with the microcanonical ensemble, which is the simplest ensemble, and then we will move to the canonical ensemble, which is more general and more useful for practical applications. Finally, we will briefly discuss quantum statistical mechanics, which is a generalization of classical statistical mechanics to quantum systems.

1.1 | Equal A Priori Probabilities

We begin recalling the foundamental principles of statistical mechanics, starting from a classical isolated system of N particles, with total energy E and volume V.

The **microstates** of the system are the points in the phase space of the system, which is a $6N$-dimensional space spanned by the coordinates and momenta of the particles. The **macrostates** of the system are defined by the macroscopic variables, such as energy, volume, and number of particles. The **ensemble** is a collection of microstates that are compatible with a given macrostate. The **probability distribution** of the system is a function that assigns a probability to each microstate in the ensemble.

It is useful to recall the Hamiltonian formalism, which is a powerful tool for describing the dynamics of classical systems. The Hamiltonian H is a function of the coordinates and momenta of the particles, and it encodes the energy of the system. The equations of motion for the system can be derived from the Hamiltonian using Hamilton's equations, which describe how the coordinates and momenta evolve in time. For each particle i we have

$$(q_1,\, q_2,\, q_3,\, p_1,\, p_2,\, p_3) \in \mathcal{M},$$

where $\mathcal{M}$ is the phase space of the system, which is a $6N$-dimensional manifold. The Hamiltonian H is a function that maps points in the phase space to real numbers, and it encodes the energy of the system

$$H = H(q_1, \dots, q_{3N},\, p_1, \dots, p_{3N}) : \mathcal{M} \to \mathbb{R}.$$

The equations of motion for the system can be derived from the Hamiltonian using Hamilton's equations, which describe how the coordinates and momenta evolve in time:

$$\begin{cases} \dot{q}_i = \dfrac{\partial H}{\partial p_i}, \\ \dot{p}_i = -\dfrac{\partial H}{\partial q_i}. \end{cases}$$

We can visualize the phase space of the system as a high-dimensional space, where each point represents a microstate of the system. The energy shell defined by the total energy E is a hypersurface in this phase space, and the trajectories of the system are represented by curves that lie on this hypersurface.

TODO: add a picture of the phase space and the energy shell

If we suppose that

1. we know positions and momenta of all particles at a given time t, thus we know the microstate of the system at time t,

2. we have an "infinite-powerful" computer that can solve the equations of motion for the system, thus we can predict the microstate of the system at any time t' given the microstate at time t,

we should be able to predict the future evolution of the system with perfect accuracy. However, in practice, we do not have access to the microstate of the system, and we do not have an infinite-powerful computer that can solve the equations of motion for the system. Specially if we think about a system with $N \sim N_A = 6 \times 10^{23}$ particles, it is impossible to know the microstate of the system, and it is impossible to solve the equations of motion for the system. We need to introduce approximations and assumptions in order to be able to make predictions about a large collection of particles.

A standard way to introduce statistical mechanics is through the so-called **postulate of equal a priori probabilities**:

Postulate 1.1 (Equal A Priori Probabilities). *Macroscopic observables can be computed defining a* ***probability distribution*** *on the phase space of the system, which assigns probabilities to all microstates, and taking the expectation value of the observable with respect to this probability distribution. In an isolated system, all microstates are equally probable, thus the probability distribution is uniform on the energy shell defined by the sub-manifold of the phase space where the energy is equal to E:*

$$\mathcal{M}_E = \{(q_1, \ldots, q_{3N}, p_1, \ldots, p_{3N}) \in \mathcal{M} : H(q_1, \ldots, q_{3N}, p_1, \ldots, p_{3N}) = E\}.$$

Let us analyze this postulate in more detail through an exemplification.

Example. Suppose we have a system of N non-interacting particles in a box of volume V, with total energy E. We are interested in computing the average particle velocity $\overline{v}_x$ along some axis x.

We should compute the time dependent function $v_x(t)$ and then take the time average of it:

$$\overline{v}_x(t) = \overline{v}_x(\{q_j(t)\}, \{p_j(t)\}) = \frac{1}{N} \sum_{j=1}^{N} \frac{v_{x,j}(t)}{m},$$

which is very complicated in general, since we have to solve the equations of motion for the system in order to find the trajectories of the particles in the phase space. However, we can use the postulate of equal a priori probabilities to compute the average particle velocity without solving the equations of motion.

The function $\overline{v}_x(\{q_j\}, \{p_j\})$ is defined on the phase space of the system, and it assigns a value to each microstate of the system. Given the knowledge on some initial microstate, the system will evolve along a curve in the fixed energy shell defined by E:

$$\mathbf{R}(t) = \{q_j(t), p_j(t)\}, \quad \overline{v}_x(t) = \overline{v}_x(\mathbf{R}(t)).$$

We can now assume the system to be at the equilibrium, thus we can take the time average of the function $\overline{v}_x(t) \sim \overline{v}_x$ over a long time interval T as

$$\overline{v}_x = \frac{1}{T} \int_0^T \mathrm{d}t \, \overline{v}_x(\mathbf{R}(t)).$$

Now we can use the postulate of equal a priori probabilities to replace the time average with an average over the energy shell defined by E:

$$\overline{v}_x = \frac{1}{|\mathcal{M}_E|} \int_{\mathcal{M}_E} \mathrm{d}\mathcal{M} \, \overline{v}_x(\{q_j\}, \{p_j\}),$$

where we are using the previous definition of the energy shell $\mathcal{M}_E$ and $|\mathcal{M}_E|$ is the volume of the energy shell in the phase space

$$|\mathcal{M}_E| = \int_{\mathcal{M}_E} \mathrm{d}\mathcal{M}.$$

Thus we have computed the average particle velocity without solving the equations of motion for the system, but using only the postulate of equal a priori probabilities: it told us that *averaging over the system trajectory is equivalent to averaging over the whole manifold at fixed energy.*

Let us now analyze the postulate of equal a priori probabilities in more detail. We have to stress some important points about it:

- In typical equilibrium situations, macroscopic quantities (such as $\overline{v}_x$ from the example) usually are constant in time only up to small fluctuations

$$\overline{v}_x(t) = \overline{v}_x + \delta v_x(t).$$

  We have previously neglected these fluctuations, but they are actually present in the system, and they can be formally removed by studying the time averages. Indeed we expect those fluctuations to be small and to occur at microscopic time scales, much smaller than the time scales of the macroscopic observables, thus they will not affect the macroscopic properties of the system. Thus measuring (i.e. averaging) over a long time interval T will remove these fluctuations:

$$\frac{1}{\tau}\int_0^\tau \mathrm{d}t\,(\overline{v}_x + \delta v_x(t)) = \frac{1}{\tau}\left(\overline{v}_x\tau + \int_0^\tau \mathrm{d}t\,\delta v_x(t)\right) = \overline{v}_x,$$

  where we have used the fact that the time average of the fluctuations is zero.

- The traditional definition of the postulate of equal a priori probabilities is simpler than the one we have given, since it is usually stated as *in an isolated system, all microstates are equally probable*, without specifying that the microstates are the points in the energy shell defined by E. We have given a more modern and precise definition of the postulate, which is more useful for our purposes, since it allows us to compute the average of macroscopic observables as an average over the energy shell in the phase space.

- In the study of dynamical systems, it is possible to mathematically formalize the problem of proving the postulate of equal a priori probabilities, and to find conditions under which it holds. This is a very active field of research, called **ergodic theory**, and in some cases one can prove its validity, while in other cases one can find counterexamples where it does not hold. There exist complicated systems for which the postulate of equal a priori probabilities does not hold, but these systems are not relevant for our purposes, thus we will assume the validity of the postulate for the rest of the course, and we will use it as a fundamental principle.

- We have assumed above that the system is at equilibrium, thus we can take the time average of the macroscopic observable over a long time interval T. If the system is not at equilibrium, then the time average of the macroscopic observable will not be constant in time, and it will depend on the initial conditions of the system. Studying how a system reaches the equilibrium (or even defining it rigorously) is non-trivial, and it is a very active field of research, called **non-equilibrium statistical mechanics** or **thermalization**. Typically the approach to equilibrium is obtained via a weak coupling among theh system and the environment, which allows the system to exchange energy and information with the environment, so that the system is not completely isolated.

With the above discussion at hand, it is useful to define the concept of **typical behaviour**. Given a function defined on the phase space (an observable) $f(\{q_j, p_j\})$ associated with some macroscopic quantity, we can define a **typical point** $\mathbf{R} = \{q_j, p_j\}$ as the point in which the function f takes a value that is close to the average value of the function over the energy shell defined by E:

$$f(\mathbf{R}) = \frac{1}{|\mathcal{M}_E|} \int_{\mathcal{M}_E} \mathrm{d}\mathcal{M}\, f(\{q_j\},\, \{p_j\}) + \delta f,$$

where δf is a small fluctuation. The postulate of equal a priori probabilities implies that the typical behaviour of the system is close to the average behaviour of the system, thus we can say that the typical point in the phase space of the system is close to the average value of the function f over the energy shell defined by E.

1.2 | Microcanonical Ensemble

The postulate of equal a priori probabilities defines the microcanonical ensemble (in general an ensemble is a collection of all possible microstates that are consistent with a given set of macroscopic constraints), which is the ensemble of microstates that are compatible with a given macrostate defined by the energy E, volume V, and number of particles N. The probability distribution of the system in the microcanonical ensemble is uniform on the energy shell defined by E, thus we can write it as

$$\rho_{mc}(\{q_j\},\{p_j\}) = \begin{cases} \dfrac{1}{|\Gamma_\Delta(E)|}, & \text{if } E \leq H(\{q_j\},\{p_j\}) \leq E+\Delta, \\ 0, & \text{otherwise,} \end{cases} \tag{1.2.1}$$

where Δ is a small energy interval that defines the width of the energy shell, and $|\Gamma_\Delta(E)|$ is the volume of the energy shell in the phase space, which can be computed by considering the normalization condition for the probability distribution

$$\int \rho_{mc}\,\mathrm{d}^{3N}q\,\mathrm{d}^{3N}p = 1 \implies \rho_{mc} = \frac{1}{\int_{\Gamma_\Delta(E)}\mathrm{d}^{3N}q\,\mathrm{d}^{3N}p} = \frac{1}{|\Gamma_\Delta(E)|}.$$

Given a function f defined on the phase space of the system, we can compute its **ensamble average**, denoted by $\langle f\rangle_{mc}$ or $\mathbb{E}_{mc}[f]$, as the average of the function over the energy shell defined by E with respect to the probability distribution of the microcanonical ensemble:

$$\langle f\rangle_{mc} = \int f(\{q_j\},\{p_j\})\rho_{mc}(\{q_j\},\{p_j\})\,\mathrm{d}^{3N}q\,\mathrm{d}^{3N}p. \tag{1.2.2}$$

A *consistency condition* for the microcanonical ensemble (or any other ensemble) is that the statistical fluctuations of macroscopic observables are small in the thermodynamic limit $N \to \infty$. This means that the variance of a macroscopic observable f should vanish in the thermodynamic limit, thus we can write

$$\sigma_f^2 = \frac{\langle f^2\rangle_{mc} - \langle f\rangle_{mc}^2}{\langle f^2\rangle_{mc}} \ll 1 \quad \text{for } N \to \infty.$$

Now we can use these conditions to further analyze this ensamble; in order to do so let us introduce the entropy S of the system, which can be used to derive the thermodynamic properties of the system, and to find the most probable configuration of the system.

1.2.1 | Entropy

As seen in previous courses, in this framework we are outlining a microscopic basis for the thermodynamic properties of the system, thus we need to introduce a microscopic definition of the entropy S of the system, which is a measure of the number of microstates available to the system at a given energy E. The **Boltzmann formula** for the entropy is given by

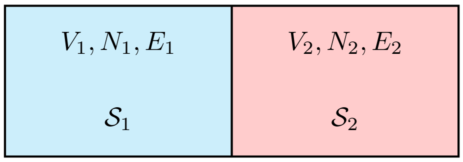
$$S(E) = k_B \log|\Gamma_\Delta(E)|, \tag{1.2.3}$$

where k_B is the Boltzmann constant, and $|\Gamma_\Delta(E)|$ is the volume of the energy shell in the phase space. The Boltzmann formula for the entropy tells us that the entropy of the system is proportional to the logarithm of the number of microstates available to the system at a given energy E. This is a fundamental result in statistical mechanics, and it allows us to derive the thermodynamic properties of the system from its microscopic properties.

We will not derive the Boltzmann formula for the entropy, but we will use it as a fundamental principle to derive the thermodynamic properties of the system. For example, we can derive a definition of temperature T of the system from the entropy S.

1.2.2 | Temperature

Let us consider two systems $\mathcal{S}_1$ and $\mathcal{S}_2$ in thermal contact such that they can exchange energy, but they are isolated from the rest of the universe. The total system is isolated, as well as the subsystems, thus we can apply the microcanonical ensemble to them.



$$\begin{cases} N = N_1 + N_2, \\ V = V_1 + V_2, \\ E = E_1 + E_2. \end{cases}$$

Thermal contact means that the two systems can exchange energy, thus the total energy E is fixed, but the energy of each subsystem E_1 and E_2 can fluctuate. The total number of particles N and the total volume V are also fixed, since there is no exchange of matter or volume between the two systems.

The total energy can now be computed from the Hamiltonian of the system, sum of the subsystems' Hamiltonians plus an interaction term

$$H(\{q_j\},\, \{p_j\}) = H_1(\{q_{1j}\},\, \{p_{1j}\}) + H_2(\{q_{2j}\},\, \{p_{2j}\}) + \delta H_{12}(\{q_{1j}\},\, \{p_{1j}\},\, \{q_{2j}\},\, \{p_{2j}\}),$$

where the last term accounts for the interaction between the two systems, which we will neglect in the following. The energy of the total system is fixed, but the energy of each subsystem can fluctuate. The probability of finding the system in a state with energy E_1 is proportional to the number of states available to the second system with energy $E_2 = E - E_1$. If we fix an energy shell of width Δ, we can write

$$E_k^{(1)} = k\Delta, \qquad E_k^{(2)} = E - E_k^{(1)} = E - k\Delta,$$

and with this "discretization" we can count the number of states available to the total system at a fixed energy E as

$$\Gamma(E) = \sum_{k=0}^{E/\Delta} \Gamma_1(E_k^{(1)})\Gamma_2(E - E_k^{(1)}),$$

where $\Gamma_k(E) = \int_{E<H_k<E+\Delta} \mathrm{d}\{q_{kf}\}\mathrm{d}\{p_{kf}\}$ is the number of states available to the k-th subsystem at energy E.

Thus we are saying that *in order to count the number of microstates at energy E we have to multiply the number of microstates of the subsystem S_1 with energy ϵ_1 to the number of microstates of S_2 at energy $\epsilon_2 = E - \epsilon_1$, then sum over all possible values of ϵ_1.*

The **most probable configuration** of the system is the one that maximizes the number of states available to the total system, which is equivalent to **maximizing the entropy** of the total system. Thus we can compute the entropy of the total system as

$$S(E) = k_B \log \left[\sum_{k=0}^{E/\Delta} \Gamma_1(E_k^{(1)})\Gamma_2(E - E_k^{(1)}) \right].$$

Now, in order to maximize it, we can take the derivative of the entropy with respect to E_1 and set it to zero: let us denote $\bar{E}_1$ the energy of the subsystem S_1 that maximizes the entropy, then $\bar{E}_2 = E - \bar{E}_1$ is the energy of the subsystem S_2 that maximizes the entropy. We have

$$\Gamma(\bar{E}^{(1)})\Gamma(\bar{E}^{(2)}) \leq \Gamma(E) \leq \frac{E}{\Delta}\Gamma(\bar{E}^{(1)})\Gamma(\bar{E}^{(2)}),$$

where the first inequality comes from the fact that $\Gamma(E)$ is a sum over all the values of E_1, thus it is greater than or equal to the single term with $E_1 = \bar{E}_1$, while the second inequality comes from the fact that there are at most E/Δ terms in the sum, thus we can upper bound it by multiplying the maximum term by the number of terms. Taking the logarithm of these inequalities we arrive at

$$S(\bar{E}^{(1)}, V_1) + S(\bar{E}^{(2)}, V_2) \leq S(E, V) \leq S(\bar{E}^{(1)}, V_1) + S(\bar{E}^{(2)}, V_2) + k_B \log\left(\frac{E}{\Delta}\right),$$

where both inequalities are saturated in the thermodynamic limit $N \to \infty$: entropy and energy are extensive quantities, thus $S(E, V) \sim N$ and $E \sim N$, so the last term in the upper bound is negligible in the thermodynamic limit

$$k_B \log\left(\frac{E}{\Delta}\right) \sim k_B \log(N) \ll N \sim S(E, V).$$

Thus we can write, in the thermodynamic limit, the entropy of the total system as the sum of the entropies of the subsystems, computed at the most probable values of the energy of the subsystems:

$$S(E,\, V) = S(\bar{E}^{(1)}, V_1) + S(\bar{E}^{(2)}, V_2).$$

The first consequence of this result is that the **entropy is an extensive quantity**, since it scales with the number of particles N in the system. The second consequence is that the *most probable configuration of the system is the one that maximizes the entropy*, which is equivalent to maximizing the number of states available to the total system. Thus we can write

$$\Gamma(E) = \Gamma_1(\bar{E}^{(1)})\Gamma_2(\bar{E}^{(2)}).$$

Therefore terms with different values of E_1 are negligible compared to the term with $E_1 = \bar{E}_1$, which is the one that maximizes the number of states available to the total system. This is a consequence of the fact that the number of states available to the total system grows exponentially with the number of particles N, thus even a small difference in energy between different configurations can lead to a huge difference in the number of states available to the total system. $\bar{E}^{(1)}$ and $\bar{E}^{(2)}$ are the most probable values of the energy of the subsystems, which are the ones that maximize the entropy of the total system: the *equilibrium energy values*. Mathematically, we can obtain $\bar{E}^{(1)}$ from the saddle-point condition:

$$\frac{\partial}{\partial \epsilon}\left[\Gamma_1(\epsilon)\Gamma_2(E-\epsilon)\right]_{\epsilon=\bar{E}^{(1)}} = 0,$$

which, expanding the derivative, can be written as

$$\left[\frac{\partial}{\partial \epsilon}\Gamma_1(\epsilon)\right]_{\epsilon=\bar{E}^{(1)}}\Gamma_2(E-\epsilon) + \Gamma_1(\epsilon)\left[-\frac{\partial}{\partial \epsilon}\Gamma_2(E-\epsilon)\right]_{\epsilon=\bar{E}^{(2)}} = 0,$$

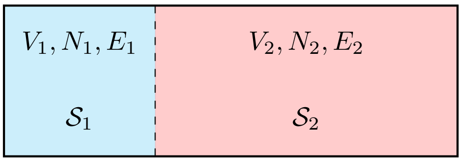
which now, dividing by $\Gamma_1(\bar{E}^{(1)})\Gamma_2(\bar{E}^{(2)})$ (in order to build the logarithmic derivative), and using the definition of entropy we arrive at

$$\left.\frac{\partial}{\partial \epsilon}S_1(\epsilon)\right|_{\epsilon=\bar{E}^{(1)}} = \left.\frac{\partial}{\partial \epsilon}S_2(\epsilon)\right|_{\epsilon=\bar{E}^{(2)}}.$$

This is a fundamental result, which tells us that at equilibrium the derivative of the entropy with respect to the energy is the same for both subsystems. This allows us to define a thermodynamic quantity which is equal in systems in thermal equilibrium, called **temperature** T (recovering the first principle of thermodynamics). The temperature is defined as the inverse of the derivative of the entropy with respect to the energy, thus we can write

$$\frac{1}{T} = \frac{\partial S}{\partial E}(E, V).$$

1.3 | Canonical Ensemble



$$\begin{cases} N = N_1 + N_2, \\ V = V_1 + V_2, \\ E = E_1 + E_2, \end{cases} \quad \text{and} \quad \begin{cases} N_2 \gg N_1, \\ V_2 \gg V_1, \\ E_2 \gg E_1. \end{cases}$$

The second system S_2 is called the *heat bath*, and it is so large that it can exchange energy with the first system S_1 without changing its thermodynamic properties. The total system is isolated, thus we can apply the microcanonical ensemble to it. The probability of finding the system in a state with energy E_1 is proportional to the number of states available to the second system with energy $E_2 = E - E_1$, which is given by $\Omega_2(E - E_1)$.

Technically speaking, we want to find the probability distribution of the subsystem S_1 when it is in contact with a heat bath S_2, knowing the total probability distribution: the total system is isolated, thus we can apply the microcanonical ensemble to it, and we know that it will be evolving with a trajectory in the phase space that is uniformly distributed on the energy shell defined by the total energy E. This means that if we analyze the evolution of the subsystem S_1 in its phase space, we will find that it will not be limited to fixed energy states, but it will be able to explore all the energy states available to it, with a probability that is proportional to the number of states available to the heat bath S_2 at the corresponding energy. we are interested in finding the **induced probability distribution** of the subsystem S_1.

Let us visualize the phase space of the total system as a high-dimensional space, where each point represents a microstate of the total system. The energy shell defined by the total energy E is a hypersurface in this phase space, and the trajectory of the total system is uniformly distributed on this hypersurface. The subsystem S_1 can be thought of as a projection of the total system onto a lower-dimensional subspace, where each point represents a microstate of the subsystem S_1. The probability distribution of the subsystem S_1 is then given by the projection of the uniform distribution on the energy shell onto this lower-dimensional subspace. Differents probabilities correspond to different energy states of the subsystem S_1, and the probability of finding the subsystem S_1 in a state with energy E_1 is proportional to the number of states available to the heat bath S_2 at energy $E - E_1$.

The idea is to fix a point in the phase space of the subsystem S_1 with energy E_1, and then count the number of points in the phase space of the heat bath S_2 that are compatible with this point, i.e. that have energy $E - E_1$. This will give us the number of states available to the heat bath S_2 at energy $E - E_1$, which is proportional to the probability of finding the subsystem S_1 in a state with energy E_1.

So we fix an energy ϵ for the subsystem S_1, and we want to find the number of states available to the heat bath S_2 at energy $E - \epsilon$. *The number of points in the phase space of S that corresponds to P is equal to the number of points in the phase space of S_2 that corresponds to $E - \epsilon$.* Thus we can write

$$\rho(\{q_f\},\,\{p_f\}) \propto \frac{\Gamma_2(E-\epsilon)}{\Gamma(E)}.$$

We know that the entropy of the heat bath S_2 is related to the number of states available to it by the Boltzmann formula (we are using $k_B = 1$ for simplicity), such that

$$\Gamma_2(E-\epsilon) = e^{S_2(E-\epsilon)} \sim e^{S_2(E)-\epsilon\frac{\partial S_2}{\partial E}\epsilon},$$

where we have used the fact that the entropy is an extensive quantity, thus we can write it as a function of the energy per particle, and we have expanded it around the total energy E of the heat bath S_2. We can now recall the temperature as a thermodynamic quantity defined as the inverse of the derivative of the entropy with respect to the energy, while reinserting the Boltzmann constant k_B for clarity, we have

$$\rho(\{q_f\}, \{p_f\}) \propto e^{-\beta\epsilon} = e^{-\frac{\epsilon}{k_B T}},$$

where $\beta = \frac{1}{k_B T}$ is the inverse temperature. Thus we have derived the Boltzmann distribution for the subsystem S_1 in contact with a heat bath S_2. The probability of finding the subsystem S_1 in a state with energy ϵ is proportional to the exponential of the negative of the energy divided by the temperature of the heat bath S_2. This is the canonical ensemble, where the subsystem S_1 is in thermal equilibrium with a heat bath S_2 at temperature T.

We can now define the partition function Z as the normalization constant for the probability distribution of the subsystem S_1, such that

$$Z = \int \frac{\mathrm{d}^{3N}q\,\mathrm{d}^{3N}p}{h^{3N}N!}\, e^{-\beta H(\{q\},\{p\})},$$

which can be used as a generating function for the thermodynamic properties of the subsystem S_1. For example, the free energy F can be computed from the partition function as

$$F = -k_B T \log Z = -\frac{1}{\beta}\log Z,$$

and the internal energy U can be computed as

$$U = -\frac{\partial}{\partial\beta}\log Z,$$

while the entropy S can be computed as

$$S = -\frac{\partial F}{\partial T} = k_B \beta^2 \frac{\partial F}{\partial \beta}.$$

Discrete System

A discrete system is a system that can only take a finite number of states, such as a spin system or a lattice gas. In this case, let us imagine a collection of N particles that can only occupy a finite number of states, let us imagine two values of spin, up and down, thus we have 2^N possible configurations of the system. The phase space of the system is now a discrete set of points, configurations (a set of sets) $\{c_j\}_{j=1}^N$, and the probability distribution of the system is defined on this discrete set of points. The partition function can be written as a sum over all possible configurations of the system, such that

$$Z = \sum_{\{c_j\}} e^{-\beta H(\{c_j\})}.$$

1.4 | Quantum Statistical Mechanics

Considering a general quantum mechanical system, we can study it with different tools, such as the hamiltonian operator, the density matrix, the partition function, etc. The Hamiltonian operator H is a self-adjoint operator that acts on the Hilbert space of the system, and it encodes the energy of the system. The density matrix ρ is a positive semi-definite operator that acts on the Hilbert space of the system, and it encodes the state of the system. The partition function Z is a scalar quantity that encodes the thermodynamic properties of the system.

$$
\begin{aligned}
\hat{H} \left| n \right\rangle &= E_n \left| n \right\rangle, \\
\left| \psi \right\rangle &\in \mathcal{H}, \\
\{\left| E_n \right\rangle\}_{n=1}^{D} \implies \left| \psi \right\rangle &= \sum_{n=1}^{D} c_n \left| E_n \right\rangle, \\
\left\langle \psi \right| \hat{A} \left| \psi \right\rangle &= \sum_{n,m} c_n^* c_m \left\langle E_n \right| \hat{A} \left| E_m \right\rangle, \\
\left| \psi(t) \right\rangle = \sum_n c_n(t) \left| E_n \right\rangle, &\quad c_n(t) = c_n e^{-i \frac{E_n}{\hbar} t} \\
\left\langle \psi(t) \right| \hat{A} \left| \psi(t) \right\rangle &= \sum_{n,m} c_n^* c_m e^{i \frac{E_n - E_m}{\hbar} t} \left\langle E_n \right| \hat{A} \left| E_m \right\rangle, \\
\implies = \frac{1}{T} \int_0^T \mathrm{d}t \, \left\langle \psi(t) \right| \hat{A} \left| \psi(t) \right\rangle &= \sum_n |c_n|^2 \left\langle E_n \right| \hat{A} \left| E_n \right\rangle.
\end{aligned}
$$

Let us report some postulates of quantum statistical mechanics:

- Postulate of equal a priori probabilities: in an isolated system, all microstates are equally probable,
$$
|\bar{c_n}|^2 = \begin{cases} \frac{1}{D}, & \text{if } E_n \in [E, E+\Delta], \\ 0, & \text{otherwise.} \end{cases}
$$
- postulate of random phases: the phases of the coefficients c_n are random variables uniformly distributed in the interval $[0, 2\pi]$, thus
$$
\frac{1}{T} \int_0^T \mathrm{d}t \, c_n^*(t) c_m(t) = \frac{1}{T} \int_0^T \mathrm{d}t \, c_n^* c_m \, e^{i \frac{E_n - E_m}{\hbar} t} = \begin{cases} |c_n|^2, & \text{if } n = m, \\ 0, & \text{otherwise.} \end{cases}
$$

The time average of an observable $\hat{A}$ is given by

$$
\overline{\left\langle \psi(t) \right| \hat{A} \left| \psi(t) \right\rangle} = \frac{1}{T} \int_0^T \mathrm{d}t \, \left\langle \psi(t) \right| \hat{A} \left| \psi(t) \right\rangle = \sum_n |c_n|^2 \left\langle E_n \right| \hat{A} \left| E_n \right\rangle.
$$

[...]

We can define the density matrix ρ as

$$
\rho = \frac{1}{Z} \sum_{E \leq E_n \leq E+\Delta} \left| E_n \right\rangle \left\langle E_n \right|,
$$

which is a positive semi-definite operator that encodes the state of the system. The expectation value of an observable $\hat{A}$ can be then computed as

$$
\langle \hat{A} \rangle = \mathrm{Tr}(\rho \hat{A}).
$$

We have to note that dealing with quantum statistical mechanics is more complicated than dealing with classical statistical mechanics, since we have to deal with operators instead of functions, and we have to deal with the non-commutativity of operators. However, the general principles of statistical mechanics still apply, and we can still define ensembles, partition functions, and thermodynamic quantities in the quantum case. The main difference is that we have to use the language of operators and Hilbert spaces instead of the language of functions and phase space.

Defining an ensemble in quantum statistical mechanics automatically defines a density matrix, which encodes the state of the system, and from which we can compute the expectation values of observables and the thermodynamic properties of the system. We are governing our set of eigenstates with a probability distribution, which is encoded in the density matrix, and we can compute the expectation values of observables and the thermodynamic properties of the system from this density matrix.

For a quantum system at thermal equilibrium, we have the following identification of the thermodynamic free energy F with the partition function Z:

$$F = -k_B T \log Z,$$

which is the same relation that we have in classical statistical mechanics. These postulates are like operational tools: justify them as they are can be rally difficult and it is an actual field of research with a lot of literature, but they are very useful for computing the thermodynamic properties of quantum systems at thermal equilibrium. For us, justifying why a precise postulate workd in the actual setting will be enough.

2 Phase Transitions

Phase Transitions are fundamental phenomena in statistical physics, where a system undergoes a qualitative change in its macroscopic properties as external parameters (like temperature or pressure) are varied. These transitions can be classified into different types based on their characteristics and the nature of the change in the system. Let us start from the definition of a phase of a many-particle system, which is crucial for understanding phase transitions.

Definition 2.1 (Phase of Many-Particle System)**.** A phase of a many-particle system is a region in the parameter space (e.g., temperature, pressure) where the system exhibits distinct physical properties and behavior: it is a "type" of collective state of matter. They are defined in terms of macroscopic properties such as density, magnetization, or symmetry.

A phase transition occurs when a system undergoes an *abrupt change from one phase to another* (states) *as we smoothly vary an external parameter.* Often PTs are accompanied by a discontinuity or singularity in some thermodynamic quantity, which serves as an order parameter to characterize the transition.

A phase transition is always a result of a competition between "order" and "disorder" in the system. For example, in a ferromagnetic material, the ordered phase corresponds to a state where the magnetic moments are aligned, while the disordered phase corresponds to a state where the magnetic moments are randomly oriented. The transition between these two phases can be driven by temperature, where thermal fluctuations disrupt the order at high temperatures.

We are now going to define PTs in a mathematical framework, which will allow us to classify them into different types. To start we have to highlight how PTs are related to **singularities in the partition function** of the system at the thermodynamic limit, or equivalently in the free energy. The **free energy** is a thermodynamic potential that encodes the equilibrium properties of a system, and its behavior near a phase transition can reveal important information about the nature of the transition:

$$F = -\frac{1}{\beta} \ln Z = U - TS,$$

where Z is the partition function of the system and $\beta = 1/(k_B T)$ is the inverse temperature. A phase transition is typically associated with a non-analytic behavior of the free energy as a function of the external parameters, such as temperature or pressure. Thus PTs are defined as singularities in the thermodynamic limit of the partition function Z or equivalently of the free energy F.

Definition 2.2 (Phase Transition)**.** A phase transition is a non-analytic behavior of the free energy F as a function of external parameters (e.g., temperature, pressure) in the thermodynamic limit. This non-analyticity manifests as a discontinuity or singularity in some thermodynamic quantity, which serves as an order parameter to characterize the transition.

Example (Finite system and phase transitions). If we consider a finite set of spins, the partition function Z is a finite sum of exponentials, which is an analytic function of the parameters:

$$Z = \sum_{\{\text{conf}\}} e^{-\beta E[\text{conf}]}.$$

Therefore, in a finite system, there can be no true phase transition, as the free energy will always be analytic. However, as we take the thermodynamic limit (i.e., let the number of spins go to infinity), the partition function can develop singularities, leading to non-analytic behavior in the free energy and thus allowing for phase transitions to occur.

In physics we never encounter real infinite systems, but we can approximate them by considering large systems and taking the thermodynamic limit. This is a common approach in statistical mechanics to study phase transitions, as it allows us to capture the essential features of the transition while avoiding the complications that arise from finite-size effects.

2.1 | Classification of Phase Transitions

Phase transitions can be classified into different types based on their characteristics and the nature of the change in the system. One of the first classifications was proposed by *Paul Ehrenfest*, who categorized phase transitions based on the order of the derivative of the free energy that exhibits a discontinuity at the transition point. Today, we have a more refined classification, which is based on the concept of order parameters and symmetry breaking.

Phase transitions can be classified depending on the type of singularity in the free energy. We can distinguish between first-order and continuous (second-order) phase transitions:

- **First-order phase transitions** are characterized by a discontinuity in the first derivative of the free energy with respect to some external parameter (e.g., temperature). This means that there is a **latent heat** associated with the transition, and the order parameter changes discontinuously.[1] Examples include the liquid-gas transition and the melting of a solid.

- **Continuous (second-order) phase transitions** are characterized by a continuous first derivative of the free energy, but a discontinuity in the second or higher derivatives. In this case, there is **no latent heat**, and the order parameter changes continuously. Examples include the ferromagnetic transition in the Ising model and the superconducting transition in certain materials. These are the most interesting PTs and will drive most of our discussion in the following chapters.

The latter group of PTs is also called **critical phenomena**, and they are associated with the emergence of long-range correlations and *scale invariance* at the critical point. The study of critical phenomena has led to the development of powerful theoretical tools, such as the *renormalization group*, which allows us to understand the universal behavior of systems near criticality. Another important phenomenon associated with continuous phase transitions is the concept of **spontaneous symmetry breaking**. In many cases, the ordered phase of a system exhibits a lower symmetry than the disordered phase. This spontaneous symmetry breaking is a key feature of many continuous phase transitions and plays a crucial role in determining the properties of the system near criticality.

2.1.1 | Phase Diagram of Water

Looking at the phase diagram of water, we can see that there are three main phases: solid (ice), liquid (water), and gas (vapor). The **phase boundaries** between these phases are represented by lines in the pressure-temperature (p-T) diagram, also called coexistence lines. The point where the liquid-gas phase boundary ends is called the **critical point**, which is a second-order phase transition (only "crossing" the specific point). Over this point, the properties of the liquid and gas phases become indistinguishable, suggesting that the two phases are not fundamentally different, and the system exhibits critical phenomena such as large fluctuations and long-range correlations.

[1] This happens because we can express order parameters as derivatives of the free energy, which is continuous but has discontinuous first derivative.

Moving along the phase boundaries, we can observe first-order phase transitions, where the system undergoes a discontinuous change in density and other properties. For example, when water freezes into ice or boils into vapor, there is a latent heat associated with the transition, and the order parameter (density) changes discontinuously. The first derivative of the free energy with respect to temperature (which is related to the entropy) and pressure (which is related to the volume) will show a discontinuity at these transitions, confirming their first-order nature.

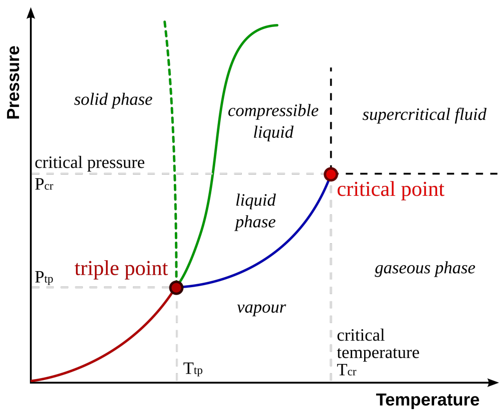
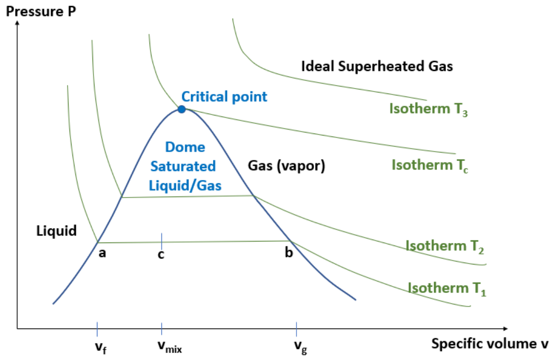


Figure 2.1: Phase diagram of water in the pressure-temperature (p-T) plane. The lines represent the phase boundaries between solid, liquid, and gas phases, with the critical point marking the end of the liquid-gas phase boundary.

Decreasing the pressure at a fixed temperature can also lead to phase transitions. For example, if we decrease the pressure of liquid water at a constant temperature, we can reach the point where it transitions into vapor. This is another example of a first-order phase transition, as there is a discontinuity in the density and other properties of the system.

Each point in the (p-T) phase diagram corresponds to a specific value of $v = \frac{V}{N}$ specific volume, except for the points on the phase boundaries (I order transition lines), where we have a coexistence of phases and the specific volume can take on different values depending on the relative proportions of the coexisting phases. For example, along the liquid-gas phase boundary, we can have a mixture of liquid and gas phases, and the specific volume will depend on the ratio of liquid to gas in the mixture. This can be seen in the following phase diagram.

The phase diagram of water in the pressure-volume (p-v) plane shows indeed the different phases and the transitions between them, with the critical point marking the end of the liquid-gas phase boundary.



If we are below the critical temperature, at a fixed T we can observe all three phases by varying the pressure or volume: this isothermal lines form a mesoscopic domain where we can observe the coexistence of phases and the transitions between them. This is a rich area of study in statistical mechanics, as it allows us to explore the behavior of systems near criticality and understand the underlying mechanisms driving phase transitions.

Figure 2.2: Phase diagram of a pure substance in the pressure-volume (p-v) plane. The lines represent the phase boundaries between different phases, with the critical point marking the end of the liquid-gas phase boundary.

The isotherms in the previous diagram are not to be confused with the coexistence lines in the p-T diagram, which represent the phase boundaries. In the p-v diagram, the isotherms represent the behavior of the system at a fixed temperature as we vary the pressure and volume, and we encounter phase I order phase transitions as we cross the coexistence region (“dome” of saturated liquid/gas). This region is characterized by a coexistence of liquid and gas phases, where the specific volume can take on different values depending on the relative proportions of the coexisting phases

$$\rho_g \sim \frac{1}{v_g}, \quad \rho_l \sim \frac{1}{v_l}, \quad \text{at } T < T_c.$$

The critical point marks the end of this coexistence region, where the properties of the liquid and gas phases become indistinguishable.

The full phase diagram of water in the pressure-volume-temperature (p-v-T) space provides a comprehensive view of the different phases and the transitions between them. It shows how the phases of water change as we vary both temperature and pressure, with the critical point marking the end of the liquid-gas phase boundary.

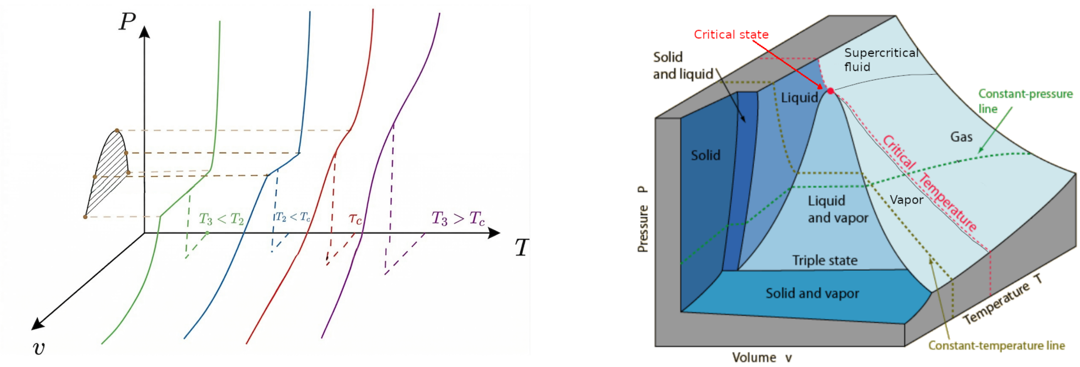


Figure 2.3: Phase diagram of water in the pressure-volume-temperature (p-v-T) space. On the right, the surfaces represent the different phases of water, with the critical point marking the end of the liquid-gas phase boundary. On the left, different isotherms are shown, illustrating how the phases change as we vary temperature and pressure (only near the critical point).

Transitions may be direction dependent in the phase space, meaning that the nature of the transition can change depending on the path taken in the parameter space. At the critical point, the system exhibits critical phenomena such as large fluctuations and long-range correlations. The properties of the system near the critical point can be described by critical exponents, which characterize how various physical quantities diverge or vanish as the critical point is approached. These critical exponents are universal, meaning that they depend only on the dimensionality of the system and the symmetry of the order parameter, rather than on the specific details of the microscopic interactions. This universality is a key feature of critical phenomena and allows us to classify different systems into universality classes based on their critical behavior.

Phase transitions are well defined when described with a curve in the parameter space, but we can also have transitions that occur at a single point, such as the critical point in the phase diagram of water. In this case, the transition is not associated with a curve but rather with a specific set of conditions (temperature and pressure) where the properties of the system change dramatically.

We are only going to discuss PTs which occurs with a smooth variation of the parameters, but there are also PTs that can occur with a sudden change in the parameters, such as a quench or a rapid cooling. These non-equilibrium phase transitions can lead to interesting phenomena such as the formation of topological defects and the emergence of new phases that are not present in equilibrium conditions. The study of non-equilibrium phase transitions is an active area of research in statistical mechanics and condensed matter physics.

Remark (Critical Opalescence). *The* ***milky appearence*** *(or critical opalescence) of a system near the critical point is a direct consequence of the divergence of the correlation length ξ (for example in a liquid-gas mixture). As the correlation length increases, fluctuations in the density or order parameter occur over larger and larger distances, leading to the scattering of light and giving rise to a milky or cloudy appearance. This phenomenon is a visual manifestation of the critical behavior of the system and is often observed in fluids near their critical point, such as in the case of water at its critical temperature and pressure.*

2.1.2 | Magnetic System

In this setting, we are considering a universal magnetic system, which can be described by a Hamiltonian that includes interactions between magnetic moments (spins) and an external magnetic field. The behavior of the system can be analyzed in terms of its phase transitions, which are characterized by changes in the order parameter (magnetization) as we vary the temperature and the external magnetic field.

We have few control parameters over the system: the temperature T and the external magnetic field h. The phase diagram of the magnetic system can be represented in the $T-h$ plane, where we can identify different phases based on the behavior of the magnetization.

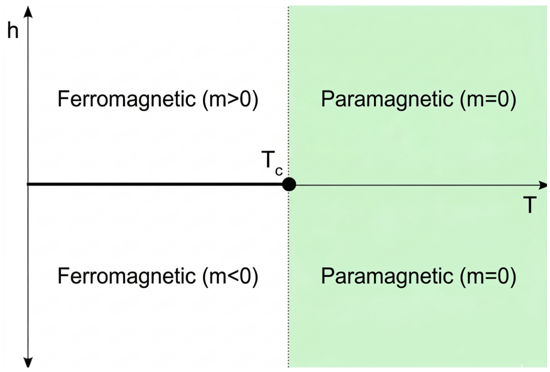


Figure 2.4: Phase diagram of a magnetic system in the (h-T) plane, showing the ferromagnetic phase at low temperatures, cut in half by the first-order phase transition line, which ends at the critical point. Over the critical point, the system is in the paramagnetic phase.

- At high temperatures and low magnetic fields, the system is in a disordered phase where the magnetization is zero: **paramagnetic phase**.
- At low temperatures and high magnetic fields, the system is in an ordered phase where the magnetization is non-zero: **ferromagnetic phase**.
- The transition between these two phases can be either first-order or second-order, depending on the specific interactions in the system. If we cross the phase buondary under the critical point, we observe a first-order phase transition, while if we cross the critical point, we observe a second-order phase transition. Over it we have only the paramagnetic phase, which is a disordered phase with zero magnetization.
- At the critical point, the system exhibits critical phenomena such as large fluctuations in magnetization and long-range correlations. The properties of the system near the critical point

can be described by critical exponents, which characterize how various physical quantities diverge or vanish as the critical point is approached.

Introducing the **magnetization** as the order parameter of the system, we can analyze how it changes across the phase transitions.

Definition 2.3 (Order Parameter)**.** An order parameter is a quantity that serves as a measure of the degree of order in a system and can be used to characterize different phases. It typically takes on a non-zero value in the ordered phase and vanishes in the disordered phase. It is expressed generally as

$$m(T) = \frac{1}{V} \lim_{h \to 0} \langle M(h,T) \rangle,$$

where the notation is deliberately describing the case of magnetization, but we can interpret more generally as the order parameter of a system, which can be defined in terms of the expectation value of some observable that characterizes the order in the system.

Thus we interpret in general as

$$\phi(T) = \frac{1}{V} \lim_{h \to 0} \langle O(h,T) \rangle,$$

where $O(h, T)$ is an observable that characterizes the order in the system, and V is the volume of the system, h is an external field that couples to the order parameter, and the limit $h \to 0$ ensures that we are measuring the spontaneous magnetization (or order parameter) in the absence of an external field.

In the paramagnetic phase, the magnetization is zero, while in the ferromagnetic phase, it takes on a non-zero value. The behavior of the magnetization near the critical point can be described by a power law, with a critical exponent that characterizes how the magnetization vanishes as we approach the critical point from the ordered phase.

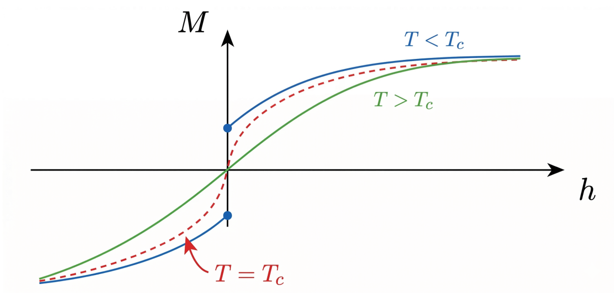


Figure 2.5: Phase diagram of a magnetic system in the (M-h) plane, showing the behavior of the magnetization as an order parameter as a function of the external magnetic field at different temperatures. The coexistence line at the critical temperature is shown.

This is equivalent to the jump of the volume in the liquid-gas transition (p-v water diagram), which is the order parameter of that system. Indeed, the red line in the previous figure represents the coexistence line, where the system can exist in both phases. The magnetization serves as a measure of the degree of order in the magnetic system, and it is associated to a first derivative of the free energy, which is discontinuous at a first-order phase transition and continuous at a second-order phase transition.

At any temperature below the critical temperature, we can observe a first-order phase transition by varying the external magnetic field. As we increase the magnetic field from zero, the system transitions from the paramagnetic phase to the ferromagnetic phase, with a discontinuous change in magnetization. This is a first-order phase transition, as there is a latent heat associated with the transition and the order parameter (magnetization) changes discontinuously.

The critical temperature separates two phases with different symmetries: the paramagnetic phase has a higher symmetry (rotational symmetry) than the ferromagnetic phase (which has a lower symmetry due to the alignment of spins). This is an example of spontaneous symmetry breaking, where the ordered phase exhibits a lower symmetry than the disordered phase. The study of this magnetic system and its phase transitions provides valuable insights into the behavior of many other systems that exhibit similar phenomena, such as superconductors and liquid crystals.

2.1.3 | Critical Behaviour

When the order parameter approaches zero continuously at the critical point, we can describe its behavior using a power law:

$$m(T) = \begin{cases} 0 & \text{for } T > T_c, \\ (T_c - T)^\beta & \text{for } T < T_c, \end{cases}$$

where β is the critical exponent that characterizes how the magnetization vanishes as we approach the critical point from the ordered phase. The value of β depends on the universality class of the system, which is determined by factors such as the dimensionality of the system and the symmetry of the order parameter. For example, in the three-dimensional Ising model, which is a prototypical model for ferromagnetic phase transitions, the critical exponent β is approximately 0.326.

The surprising thing is that if we compute the same exponent for a completely different system, such as the liquid-gas transition, using some difference in densities as the order parameter, we find the same value of β:

$$\begin{cases} |\rho_l - \rho_c| \sim (T - T_c)^\beta, \\ |\rho_c - \rho_g| \sim (T - T_c)^\beta, \end{cases} \quad \beta \approx 0.326,$$

where ρ_l and ρ_g are the densities of the liquid and gas phases, respectively, ρ_c at the critical point, and the critical exponent β is exactely the same as in the magnetic system, despite the fact that these systems have different microscopic details and interactions.

This is a manifestation of the **universality of critical phenomena**, which states that systems with different microscopic details can exhibit the same critical behavior near the phase transition, as long as they belong to the same universality class. This universality allows us to classify different systems based on their critical exponents and understand their behavior near criticality without needing to know the specific details of their interactions.

2.1.4 | Critical Exponents and Universality Classes

The critical exponents are a set of numbers that characterize the behavior of physical quantities near the critical point. They describe how various properties of the system diverge or vanish as the critical point is approached. Some of the most common critical exponents include:

- β: describes the behavior of the **order parameter** *as a function of temperature* near the critical point, $m(T) \sim (T_c - T)^\beta$ for $T < T_c$. We have already discussed this exponent in the context of the magnetization and the liquid-gas transition, where it characterizes how the order parameter vanishes as we approach the critical point from the ordered phase.

- α: describes the divergence of the specific heat (heat capacity) at constant volume and zero external field as a function of temperature near the critical point,

$$C_V \sim |T - T_c|^{-\alpha},$$

and the heat capacity is defined as the second derivative of the free energy with respect to temperature:

$$C_V = \frac{\partial U}{\partial T} = \frac{\partial}{\partial T}(F + TS) = T\frac{\partial S}{\partial T} = -T\frac{\partial^2 F}{\partial T^2}.$$

If we denote by $c_{\pm}(T, h = 0)$ the heat capacity at zero external field above/below the critical point ($t = \frac{T-T_c}{T_c}$), we can express the critical behavior of the heat capacity as

$$c_{\pm}(t, h = 0) \sim |t|^{-\alpha_{\pm}},$$

where $\alpha_{\pm}$ are the critical exponents for the heat capacity, in principle different above and below the critical point.

- δ: describes the behavior of the **order parameter** *as a function of the external field* at the critical temperature,

  $$m(h) \sim h^{1/\delta} \quad \text{at } T = T_c,$$

  where h is the external magnetic field. This means that the magnetization, along the critical isotherm, behaves as a power law in the external field, going to zero as the field goes to zero with the previus dependence on δ.

- γ: describes the divergence of the *zero-field* **susceptibility** *as a function of temperature*:

  $$\chi \sim |T - T_c|^{-\gamma},$$

  where the susceptibility χ is defined as the derivative of the order parameter with respect to the external field:

  $$\chi_{\pm} = \chi(t \to 0^{\pm}) = \left(\frac{\partial m}{\partial h}\right)_{h=0}, \quad \chi_{\pm} \sim |t|^{-\gamma_{\pm}},$$

  with $t = (T - T_c)/T_c$ being again the reduced temperature. The susceptibility measures how responsive the system is to changes in the external field, and its divergence at the critical point indicates that even a small change in the magnetic field can lead to a large change in the magnetization. This is a hallmark of criticality and is associated with the emergence of long-range correlations in the system.

Correlation Length and Universality

Before continuing with the definitions of ν and η, we have to introduce the concept of **correlation length**, which is a measure of how far correlations in the system extend. Near the critical point, the correlation length diverges, indicating that fluctuations occur over increasingly large distances. Let us introduce it through a little model: let us consider a system of spins on a lattice, where each spin s_i can take on values of +1 or -1. The magnetization m is defined as the average spin per site:

$$m = \frac{1}{N}\sum_{i=1}^{N}\langle s_i\rangle,$$

where the angle brackets denote the thermal average:

$$\langle s_i\rangle = \mathbb{E}_c\left[s_i\right] = \frac{1}{Z}\sum_{\{s_k\}} s_i e^{-\beta H(\{s_k\}, h)},$$

with Z being the partition function and H the Hamiltonian of the system. The correlation function $G(r)$ is defined as the average relative alignment of spins, i.e. the average product of spins at a distance r:

$$G_{ij}^{(2)} = \langle s_i s_j \rangle = \frac{1}{Z} \sum_{\{s_k\}} s_i s_j e^{-\beta H(\{s_k\},h)} = G^{(2)}(\mathbf{r}), \quad \text{where } \mathbf{r} = |\mathbf{r}_i - \mathbf{r}_j|.$$

Since $G_{ij}^{(2)}$ depends only on the distance between spins for translational invariance, it depends only on the distance r. We can then define the **connected correlation function**, which measures the correlations between fluctuations in the system:

$$G_c^{(2)}(\mathbf{r}) = \langle (s_i - \langle s_i \rangle)(s_j - \langle s_j \rangle) \rangle = \langle s_i \rangle \langle s_j \rangle - \langle s_i \rangle \langle s_j \rangle.$$

The correlation length ξ is defined as the characteristic length scale over which correlations decay in the system, away from the critical point:

$$G_c^{(2)}(\mathbf{r}) \sim e^{-r/\xi} \quad \text{for } r \gg \xi.$$

Near the critical point, the correlation function typically decays as a power law, which can be described by the critical exponent η. Let us introduce two more critical exponents describing the behavior of the correlation function and the correlation length near the critical point:

- ν: describes the divergence of the correlation length, $\xi_\pm \sim |T - T_c|^{-\nu_\pm}$, with $\nu_\pm > 0$.
- η: describes the behavior of the correlation function at the critical point, $G(r) \sim r^{-(d-2+\eta)}$ at $T = T_c$, where d is the dimensionality of the system.

Remark (Critical Opalescence and Correlation Length). *Going back to the phenomenon of critical opalescence, we can understand it better after having introduced the correlation length. Since the wavelength of visible light is much larger than the microscopic length scales of the system (i.e. atomic distances), the scattering of light must be related to density fluctuations that occur over large distances. Near the critical point, the correlation length diverges, which means that fluctuations occur over increasingly large distances. This leads to a significant scattering of light, giving rise to the milky or cloudy appearance known as critical opalescence. When the correlation length diverges, the system becomes increasingly sensitive to fluctuations, and exhibits macroscopic collective behavior that allow us to ignore the microscopic details of the system and focus on the universal properties that emerge near criticality.*

Now we are ready to define and discuss the concept of universality classes. Systems that share the same critical exponents and scaling behavior near the critical point are said to belong to the same universality class. The universality class of a system is determined by factors such as the dimensionality of the system, the symmetry of the order parameter, and the range of interactions between particles.

Definition 2.4 (Universality Class)**.** A universality class is a category of systems that share the same critical exponents and scaling behavior near the critical point, regardless of their microscopic details. Systems that belong to the same universality class exhibit the same critical behavior and can be described by the same effective field theory near the critical point. The universality class of a system is determined by factors such as the dimensionality of the system, the symmetry of the order parameter, and the range of interactions between particles.

Let us now make some observations about the critical behavior of systems near phase transitions, which will be important for our understanding of critical phenomena and the renormalization group.

Remark. *There is a connection between the divergence of the correlation length the response funcion. The connection relies on the fact that near the critical point, the system exhibits long-range correlations, which means that fluctuations occur over increasingly large distances. This leads to a divergence in the correlation length ξ, which in turn causes the response function (such as susceptibility) to diverge as well.*

TODO: *add reference to Kardar - stat ph of fields, 1.4*

The response function measures indeed how the system responds to external perturbations, and its divergence at the critical point indicates that even a small change in the external field can lead to a large change in the order parameter, which is a hallmark of criticality.

Remark. *Second order phase transitions are also called* ***continuous phase transitions*** *because the order parameter changes continuously across the transition, and there is no latent heat associated with the transition. As we will see, these PTs are associated with the* ***spontanoeus symmetry breaking****. In some cases, the symmetry is only* ***emergent****, meaning that it is not present in the microscopic description of the system but emerges as a result of the collective behavior of the system near the critical point (approximate symmetry which becomes exact only near the critical point). This emergent symmetry can play a crucial role in determining the properties of the system and its critical behavior.*

TODO: *add reference to J. Cardy - scaling and renormalization in statistical physics, 1.2*

2.2 | Mean Field Theory

TODO: reference: D. Tong - statistical field theory, 1 | D. Tong - statistical physics, 5

Mean field theory is a powerful approximation method used to study phase transitions and critical phenomena in many-body systems. The basic idea behind mean field theory is to replace the complex interactions between particles in the system with an average or "mean" field that each particle experiences. This simplification allows us to derive self-consistent equations for the order parameter and analyze the behavior of the system near the critical point.

Let us concentrate on a classical model, the Ising model, which is a prototypical model for studying ferromagnetic phase transitions.

2.2.1 | Ising Model

TODO: insert images of spins on a lattice

The Ising model consists of a lattice of spins, where each spin s_i can take on values of +1 or -1. The Hamiltonian of the Ising model is given by:

$$H = -J \sum_{\langle i,j \rangle} s_i s_j - h \sum_i s_i,$$

where J is the coupling constant that determines the strength of the interaction between neighboring spins, h is the external magnetic field, and the sum $\langle i, j \rangle$ runs over pairs of **nearest-neighbor** spins. The first term in the Hamiltonian represents the interaction between neighboring spins, which favors alignment (ferromagnetic order) when $J > 0$. The second term represents the coupling of the spins to the external magnetic field, which tends to align the spins in the direction of the field.

Definition 2.5 (Coordination Number)**.** The coordination number z of a lattice is the number of nearest neighbors that each site has. For example, in a two-dimensional square lattice, each site has 4 nearest neighbors, so the coordination number is $z = 4$. In a three-dimensional cubic lattice, each site has 6 nearest neighbors, so the coordination number is $z = 6$. The coordination number plays an important role in determining the behavior of the system and the nature of the phase transition.

Our aim is to study the equilibrium properties (where observables can effectively be calculated through thermal averaging) of the Ising model and understand the nature of the phase transition it exhibits. To do this, we will apply mean field theory to derive self-consistent equations for the magnetization (order parameter) and analyze the behavior of the system near the critical point.

Each configuration of the system is specified by the set of spin values $\{s_i\}_{i=1}^N$, and it is associated to a probability given by the Boltzmann distribution:

$$P(\{s_i\}) = \frac{1}{Z} e^{-\beta H(\{s_i\})},$$

where Z is the **partition function**, defined as the sum over all possible configurations:

$$Z = \sum_{\{s_i\}} e^{-\beta H(\{s_i\})}.$$

Example (Pair of spins). When we have only two spins, the partition function can be calculated explicitly. The Hamiltonian for two spins s_1 and s_2 is given by:

$$H = -J s_1 s_2 - h(s_1 + s_2),$$

where the sum is removed since we only have two spins. We can now compute the energy for each of the four possible configurations of the spins:

- $s_1 = +1$, $s_2 = +1$: $H = -J - 2h$.
- $s_1 = +1$, $s_2 = -1$: $H = J$.
- $s_1 = -1$, $s_2 = +1$: $H = J$.
- $s_1 = -1$, $s_2 = -1$: $H = -J + 2h$.

The partition function is then given by:

$$Z = e^{\beta J + 2\beta h} + 2e^{-\beta J} + e^{\beta J - 2\beta h}.$$

If we now wanted to compute the probability of finding the system in the configuration $s_1 = +1$, $s_2 = +1$, we would use the Boltzmann distribution:

$$P(s_1 = +1, s_2 = +1) = \frac{e^{\beta J + 2\beta h}}{Z}.$$

For the magnetization, we can compute the average spin per site:

$$M = \mathbb{E}_c[\sum_i s_i], \quad m = \frac{M}{N} = \frac{1}{N}\mathbb{E}_c[s_1 + s_2],$$

where the summations run over the configuration space.

$$m = \sum_{\langle s_i \rangle} p(\{s_i\}) \left(\frac{1}{N} \sum_i s_i \right),$$

where the sum runs over all possible configurations of the spins. In this case, we can explicitly compute the magnetization by summing over the four configurations.

If we were now to consider a system with $N = 100$, we would have 2^{100} possible configurations for a chain with $z = 2$, which is an astronomically large number. This makes it impossible to compute the partition function and the magnetization by explicitly summing over all configurations. This is where mean field theory comes into play, as it allows us to approximate the behavior of the system without needing to consider every possible configuration.

2.2.2 | Mean Field Approach - Euristic

Let us now apply the mean field approximation to the Ising model. The key idea is to replace the interaction term in the Hamiltonian with an average or "mean" field that each spin experiences. This allows us to derive a self-consistent equation for the magnetization.

It is a *euristic* approach, which is not exact but provides a good approximation, remaining not systematic, not rigorous and hard to fully justify. We will see how it can be derived from a more rigorous approach, which is the **Landau-Ginzburg effective field theory**.

There are many versions of this approach (different implementations of the approximation), but the idea is almost the same: we want to replace some microscopic degrees of freedom with an average field, which is the mean field. The mean field is defined as the average value of the spin at a given site:

Let us rewrite the Hamiltonian of the Ising model

$$H = -J \sum_{\langle s_i, s_j \rangle} s_i s_j - h \sum_i s_i.$$

If we now consider the case in which the coupling constant J is positive ($J > 0$ which corresponds to a ferromagnetic interaction), and a null external magnetic field ($h = 0$), the Hamiltonian simplifies to:

$$H = -J \sum_{\langle s_i, s_j \rangle} s_i s_j.$$

In this case the system exhibits a phase transition at a critical temperature T_c, below which the spins tend to align and the system enters an ordered ferromagnetic phase, while above T_c the spins are disordered and the system is in a paramagnetic phase.

Being interested in the behavior of the magnetization, we can express it as a thermal average in the canonical ensemble:

$$\mathbb{E}_c \left[\frac{1}{N} \sum_i s_i \right] = \frac{1}{N} \sum_{\{s_i\}} \left(\sum_i s_i \right) \frac{e^{-\beta H(\{s_i\})}}{Z},$$

where the sum runs over all possible configurations of the spins. The mean field approximation consists in building a term which encodes the average effect of the neighboring spins on a given spin: it is natural to think about building this average field from the magnetization, so let us analyze it in more detail. The magnetization can assume the following values:

$$M \in [-N, \, -N+2, \, -N+4, \, \ldots, \, +N],$$

where the values change by 2 because each spin can flip from +1 to -1 or vice versa. The magnetization per site is then given by:

$$m = \frac{M}{N} \in \left[-1, \, -1 + \frac{2}{N}, \, -1 + \frac{4}{N}, \, \ldots, \, +1 \right].$$

The average field we are building has to be proportional to the magnetization m, which serves as a measure of the degree of order in the system. Thus we can express the sum over configurations of the spins as a sum over the magnetization M and then a sum over all configurations that correspond to that magnetization:

$$\sum_{\{s_i\}_{i=1}^N} [\ldots] = \sum_{M=-N}^{N} \sum_{\{s_i\}_{i=1}^N} \delta \left(\sum_i s_i - M \right) [\ldots],$$

where we have introduced a delta function to enforce the constraint that the sum of the spins equals the magnetization M. This allows us to rewrite the partition function as a sum over the magnetization, but let's understand better the meaning of this delta function through an example.

Example ($N = 2$). If we have only two spins, the possible configurations and their corresponding magnetizations are:

- A: $s_1 = +1$, $s_2 = +1$: $M = 2$.
- B: $s_1 = +1$, $s_2 = -1$: $M = 0$.
- C: $s_1 = -1$, $s_2 = +1$: $M = 0$.
- D: $s_1 = -1$, $s_2 = -1$: $M = -2$.

The delta function $\delta\left(\sum_i s_i - M\right)$ ensures that we only sum over configurations that correspond to a specific magnetization M.

Let us analyze the contribution of each configuration to the partition function: starting from a sum over all configurations, we get

$$Z = \sum_{\{s_i\}} e^{-\beta H(\{s_i\})} = e^{-\beta H(A)} + e^{-\beta H(B)} + e^{-\beta H(C)} + e^{-\beta H(D)},$$

where $H(A)$, $H(B)$, $H(C)$, and $H(D)$ are the Hamiltonian evaluated at each configuration: considering the case $h = 0$ for simplicity, we have

$$H(A) = -J, \quad H(B) = J, \quad H(C) = J, \quad H(D) = -J.$$

Thus the partition function can be rewritten as

$$Z = 2(e^{\beta J} + e^{-\beta J}).$$

We now want to get the same result using the sum over magnetization, which is given by

$$Z = \sum_{M=-2}^{2} \sum_{\{s_i\}} \delta\left(\sum_i s_i - M\right) e^{-\beta H(\{s_i\})} = \sum_{M=-2}^{2} e^{-\beta H(M)} \sum_{\{s_i\}} \delta\left(\sum_i s_i - M\right) = 2(e^{\beta J} + e^{-\beta J}),$$

where we have used the fact that the Hamiltonian depends only on the magnetization M and not on the specific configuration of the spins, and that the sum over configurations for each magnetization value is just the number of configurations with that value of magnetization. This example illustrates how the sum over configurations can be rewritten as a sum over magnetization, with the delta function ensuring that we only consider configurations that correspond to a specific magnetization.

Now we can implement the mean field approximation by replacing the interaction term in the Hamiltonian with an average field that each spin experiences. This average field is proportional to the magnetization m, which serves as a measure of the degree of order in the system: in a fixed magnetization sector (m-sector) we can replace the interaction term with an effective field that depends on the magnetization (reinserting the external magnetic field h):

$$H_{mf}(\{s_i\}, m) = -J \sum_{\langle i,j \rangle} s_i\, m - h \sum_i s_i,$$

where we are summing over single sites (only s_i appears in the sum), but we kept the notation of the sum over nearest neighbors to indicate that the effective field is proportional to the number of nearest neighbors (coordination number z) and the magnetization m. This is the point where we have to accept that this method works even if it is hard to justify: there exist some ad-hoc argument, but it would be meaningless to use those since we will anyway present the Landau-Ginzburg effective field theory in the following sections, which then will make us able to justify this approximation in a systematic way.

Remark. *The hamiltonian H_{mf} is independent of the specific configuration of the spins $\{s_i\}$, as long as*

$$\sum_i s_i = M,$$

So that we have efficiently replaced the interaction term with an average field that depends only on the magnetization m. This allows us to derive a self-consistent equation for the magnetization and analyze the behavior of the system near the critical point without needing to consider every possible configuration of the spins.

If we continue the computation of the mean field hamiltonian we get

$$H_{mf}(\{s_i\}, m) = -Jm \sum_{\langle i,j \rangle} s_i - h \sum_i s_i = -(Jzm + h) \sum_i s_i = -(Jzm + h)Nm,$$

where we have used the fact that the sum over the spins is equal to the magnetization $M = Nm$ and the sum over the nearest neighbors gives a factor of the coordination number $\frac{Nz}{2}$.[2] This mean field Hamiltonian describes a system of non-interacting spins in an *effective magnetic field* that depends on the magnetization m and the external field h.

The average magnetization per spin in the canonical ensamble can now be performed easily applying the mean field approximation:

$$\begin{aligned} \mathbb{E}_c \left[\frac{1}{N} \sum_i s_i \right] &= \sum_{m \in [-1, \dots, +1]} m \sum_{\{s_i\}} \delta \left(\sum_i s_i - Nm \right) \frac{e^{-\beta H_{mf}(m)}}{Z} \\ &= \frac{1}{Z} \sum_m m e^{-\beta H_{mf}(m)} \left[\sum_{\{s_i\}} \delta \left(\sum_i s_i - Nm \right) \right], \end{aligned}$$

where the last term identifieas a sum over all the configurations of the spins that correspond to a given magnetization m. Note that a configuration of $N_\uparrow$ of up spins and $N_\downarrow$ of down spins corresponds to a magnetization

$$m = \frac{N_\uparrow - N_\downarrow}{N} = \frac{2N_\uparrow - N}{N} = 2\frac{N_\uparrow}{N} - 1.$$

The number of such configurations is given by the binomial coefficient:

$$\binom{N}{N_\uparrow} = \frac{N!}{N_\uparrow!(N - N_\uparrow)!} = \Omega(N_\uparrow),$$

which we can approximate using Stirling's formula for large N:

$$\log(\Omega(N_\uparrow)) \approx N \log N - N_\uparrow \log N_\uparrow - (N - N_\uparrow) \log(N - N_\uparrow),$$

which after some algebraic manipulations can be rewritten as

$$\log(\Omega(N_\uparrow)) \approx N \left[\log(2) - \frac{(1+m)}{2} \log(1+m) - \frac{(1-m)}{2} \log(1-m) \right].$$

Thus we have found an expression for the number of configurations that correspond to a given magnetization m, which can be used to compute the final form of the magnetization:

$$\mathbb{E}_c \left[\frac{1}{N} \sum_i s_i \right] = \frac{1}{Z} \sum_m m e^{-\beta N f(m)} = \frac{1}{Z} \sum_m m e^{-\beta H_{mf}(m)} \Omega(m),$$

where we have defined the function $f(m)$ as

$$f(m) = -hm - \frac{Jzm^2}{2} - T \left[\log(2) - \frac{(1+m)}{2} \log(1+m) - \frac{(1-m)}{2} \log(1-m) \right],$$

which can be interpreted as a **free energy density** (per site) that depends on the magnetization m. Indeed we have the last sum over the magnetization m getting dominated by the minimum of

[2] When we count the same site for each of its n.n., we are counting each s_i a number of z times, but we have to divide by 2 to avoid double-counting. Thus the total is $N \cdot \frac{z}{2}$.

the function $f(m)$ in the thermodynamic limit $N \to \infty$, while the other terms get suppressed by the factor $e^{-\beta N f(m)}$.[3] Thus the average magnetization can be approximated as

$$\mathbb{E}_c\left[\frac{1}{N}\sum_i s_i\right] \approx \frac{1}{e^{-\beta N f(m^*)}}\left(m^* e^{-\beta N f(m^*)}\right) = m^*, \qquad \left.\frac{\partial f(m)}{\partial m}\right|_{m^*} = 0,$$

where m^* is the value of m that minimizes the function $f(m)$. This is known as a *saddle point approximation*, which is a common technique used in statistical mechanics to evaluate integrals or sums that are dominated by the contribution of a single point (the minimum of the free energy in this case). We can use the same approximation to compute the partition function Z as well, which is given by

TODO: Add a figure of a parabula with the minimum at m-star.

$$Z = \sum_m e^{-\beta N f(m)} \approx e^{-\beta N f(m^*)}.$$

Self-consistency relation. Now computing the derivative of $f(m)$ with respect to m and setting it to zero, we get the self-consistent equation for the magnetization:

$$\beta(h + Jzm) = \frac{1}{2}\log\left(\frac{1+m}{1-m}\right) = \tanh^{-1}(m),$$

where the last step is obtaines after some algebraic manipulations. This equation can be rewritten as

$$m = \tanh\left(\beta h_{eff}\right), \quad h_{eff} = h + Jzm,$$

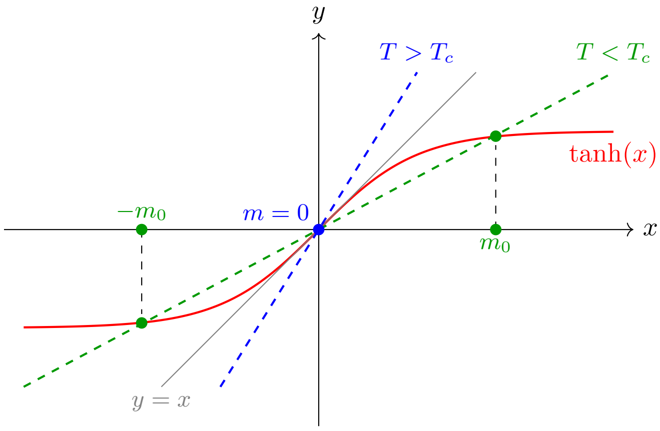
where h_{eff} is the effective magnetic field that each spin experiences due to the mean field approximation. This self-consistent equation can be solved numerically to find the magnetization m as a function of temperature T and external field h, allowing us to analyze the behavior of the system near the critical point and understand the nature of the phase transition. The effective field h_{eff} captures the influence of the neighboring spins on a given spin, and the self-consistent equation reflects the fact that the magnetization m depends on itself through the effective field. This is a hallmark of mean field theory, where the order parameter (magnetization) is determined by a self-consistent equation that takes into account the average effect of the interactions in the system.

2.2.3 | Self-Consistency and Phase Transitions

In order to solve the self-consistent equation, we can analyze it graphically by plotting the function m on the left-hand side and the function $\tanh\left(\beta h_{eff}\right)$ on the right-hand side as functions of m. The points where these two functions intersect correspond to the solutions of the self-consistent equation. By varying the temperature T and the external field h, we can observe how the number and nature of the solutions change, which provides insights into the phase transition and critical behavior of the system. Let us start from the case $h = 0$:

TODO: fix the plot

[3] Since we always have to consider an enormously large N (the real world is at the thermodynamic limit), even small variations in $f(m)$ will be exponentially suppressed.



The graphical solution shows that for $T > T_c$ the only intersection is at the origin, yielding $m = 0$: this corresponds to the **paramagnetic phase**. For $T < T_c$, instead, the line intersects the curve at two symmetric non-zero points, corresponding to two stable ordered states with opposite magnetization: this is the **ferromagnetic phase**.

- For $T > T_c$, there is only one solution at $m = 0$, which corresponds to the paramagnetic phase where the spins are disordered and there is no net magnetization.
- For $T < T_c$, there are three solutions: one at $m = 0$ and two symmetric solutions at $m = \pm m^*$, which correspond to the ferromagnetic phase where the spins are aligned and there is a non-zero magnetization. The solution at $m = 0$ becomes unstable (not a minimum) in this regime, while the two symmetric solutions represent the stable ordered phases.
- At $T = T_c$, the solution at $m = 0$ becomes marginally stable, and the system undergoes a second-order phase transition from the paramagnetic phase to the ferromagnetic phase. The magnetization m changes continuously across the transition, and there is no latent heat associated with the transition.

The critical temperature T_c is determined by the condition that the slope of the tanh function at $m = 0$ equals 1, which gives

$$\beta_c Jz = 1 \quad \Rightarrow \quad T_c = \frac{Jz}{k_B},$$

where k_B is the Boltzmann constant. Thus the condition on the critical temperature can be expressed as $\beta_c Jz = 1$.

TODO: add plot of m vs T

We can now study the beaviour of the magnetization as a function of temperature. As we decreas the temperature T below T_c, the magnetization m increases continuously from zero, indicating the onset of ferromagnetic order. We got two symmetric solutions for the magnetization, which correspond to the two possible orientations of the spins in the ferromagnetic phase. The magnetization m serves as an order parameter that characterizes the phase transition: it is zero in the paramagnetic phase and becomes non-zero in the ferromagnetic phase, indicating the spontaneous breaking of symmetry.

We can obtain the plot of the free energy as a function of the magnetization m for different temperatures to visualize the phase transition. For $T > T_c$, the free energy has a single minimum at $m = 0$, corresponding to the stable paramagnetic phase. As we decrease the temperature below T_c, the free energy develops two symmetric minima at $m = \pm m^*$, corresponding to the stable ferromagnetic phases, while the minimum at $m = 0$ becomes a local maximum, indicating that it is no longer a stable solution. This can be seen studying the expression for the free energy density $f(m)$ that we derived earlier as $h = 0$:

$$f(m) = \ldots$$

TODO: add plot of f(m) vs m for different T

and we get then two different expressions for $T > T_c$ and $T < T_c$:

- For $T > T_c$, the free energy has a single minimum at $m = 0$, which corresponds to the stable paramagnetic phase: we have a parabula with a minimum at the origin, which is the only stable solution in this regime.
- For $T < T_c$, the free energy develops two symmetric minima at $m = \pm m^*$, which correspond to the stable ferromagnetic phases, while the minimum at $m = 0$ becomes a local maximum, indicating that it is no longer a stable solution: we have a double-well potential with two minima at $m = \pm m^*$ and a local maximum at $m = 0$. As the temperature decreases further below T_c, the minima at $m = \pm m^*$ become deeper, indicating that the ferromagnetic phases become more stable.

If we were to plot the free energy as a function of the magnetization m for different temperatures, we would see how the shape of the free energy changes across the phase transition, with the emergence of two minima below T_c and the disappearance of the minimum at $m = 0$. This visualization helps to understand the nature of the phase transition and the role of the magnetization as an order parameter.

If we now imagine to "turn on" the external magnetic field h, we would see that the symmetry between the two minima at $m = \pm m^*$ is broken, and one of the minima becomes deeper than the other, depending on the sign of h. The easier way is to consider an external field which is positive but very small in comparison to the coupling constant J, so that we can treat it as a perturbation.

TODO: add plot of f(m) vs m for different h

- For $h > 0$, the minimum at $m = +m^*$ becomes deeper than the minimum at $m = -m^*$, indicating that the ferromagnetic phase with positive magnetization is more stable than the one with negative magnetization. The system will preferentially align its spins in the direction of the external field, leading to a net positive magnetization.
- For $h < 0$, the minimum at $m = -m^*$ becomes deeper than the minimum at $m = +m^*$, indicating that the ferromagnetic phase with negative magnetization is more stable than the one with positive magnetization. The system will preferentially align its spins in the opposite direction of the external field, leading to a net negative magnetization.

This means that the external magnetic field h breaks the $\mathbb{Z}_2$ symmetry between the two ferromagnetic phases and selects one of them as the stable phase, depending on the sign of h. This is a manifestation of the fact that the external field acts as a symmetry-breaking field, which can influence the behavior of the system near the critical point and affect the nature of the phase transition, especially in the presence of a small external field where the system can exhibit critical behavior and scaling properties.

Remark (On Symmetries and Spontaneous Symmetry Breaking). *A symmetry of a system is a transformation that leaves the Hamiltonian (and so the physical properties and evolution of the system) invariant.*

In the case of the Ising model with $h = 0$, there is a $\mathbb{Z}_2$ symmetry corresponding to the transformation $s_i \to -s_i$ for all spins, which leaves the Hamiltonian unchanged:

$$H(\{s_1, s_2, \ldots, s_n\}) = H(\{-s_1, -s_2, \ldots, -s_n\}).$$

When we flip all spins, is like changing the sign of the magnetization m, but the Hamiltonian remains the same; in the graph of the free energy, this corresponds to the symmetry of the double-well potential around $m = 0$. This symmetry is a fundamental property of the system *and plays a crucial role in determining the nature of the phase transition and the behavior of the system near the critical point.*

This symmetry is ***spontaneously broken*** *as $T > T_c$, since ...*

Once the system is in one of the two ferromagnetic phases (with $m = +m^*$ or $m = -m^*$), we could expect the system to be able to transition between these two phases by flipping all the spins, since the states are completely symmetric: same energy, entropy etc. However, in the thermodynamic limit $N \to \infty$, the system becomes trapped in one of the two minima of the free energy, and it cannot transition to the other minimum due to the presence of an infinite energy barrier between them: the transition probability between the two minima approches zero as a power law, and the system remains in one of the two ferromagnetic phases.

2.2.4 | Critical Exponents

The mean field theory allows us to compute the critical exponents that characterize the behavior of the system near the critical point. By analyzing the self-consistent equation for the magnetization and the free energy, we can derive expressions for the critical exponents α, β, γ, and δ in terms of the parameters of the model. The mean field theory predicts specific values for these critical exponents, which can be compared to experimental results and more sophisticated theoretical approaches to understand the limitations and applicability of mean field theory in describing critical phenomena.

We start from the analysis of the **free energy** $f(m)$ and the magnetization m as a function of temperature T near the critical point T_c. We can expand the the free energy in a Taylor series around $m = 0$ to analyze the behavior of the system near the critical point. The expansion takes the form:

TODO: correct the expansion

$$f(m) \sim T\left[\frac{m^2}{2}\left(1 - \frac{T}{T_c}\right) + \frac{m^4}{12}T_c^3 + O(m^6)\right],$$

where the interesting feature is that the coefficient of the m^2 term changes sign at $T = T_c$, which indicates the onset of the phase transition. We can now find an analytic expression for the derivative of the free energy with respect to m and set it to zero to find the equilibrium magnetization m as a function of temperature T:

$$\frac{\partial f}{\partial m} \propto m(T - T_c) + Bm^3 + O(m^5) = 0, \quad B \sim O(1) > 0,$$

which can be solved to find the behavior of the magnetization near the critical point. For $T > T_c$, the only solution is $m = 0$, while for $T < T_c$, we have two symmetric solutions given by

$$m = \pm\sqrt{\frac{T_c - T}{B}} \sim (T_c - T)^{1/2},$$

which indicates that the magnetization m behaves as a power law near the critical point with a critical exponent

$$m \propto (T_c - T)^{\beta}, \quad \beta = \frac{1}{2}.$$

This critical exponent β characterizes the behavior of the magnetization as the system approaches the critical temperature from below, and it is a key quantity in understanding the nature of the phase transition.

We can now analyze the behavior of the **magnetic susceptibility** χ near the critical point. The magnetic susceptibility is defined as the derivative of the magnetization with respect to the external magnetic field h at $h = 0$:

$$\chi = N \frac{\partial m}{\partial h}\bigg|_{h=0} .$$

Recovering the expression for the magnetization

$$m = \tanh\left(\beta(h + Jzm)\right),$$

we know that near the critical point (approached from above)T_c, the magnetization m is small, so we can expand the tanh function to first order in m:

$$m \approx \beta(h + Jzm).$$

Thus the magnetic susceptibility can be computed as

$$\chi = \frac{N\beta}{1 - \beta Jz},$$

which diverges as T approaches T_c from above, since $\beta Jz = 1$ at $T = T_c$. The critical exponent γ characterizes the divergence of the magnetic susceptibility near the critical point, and it can be defined as

$$\chi \propto |T - T_c|^{-\gamma}, \quad \gamma = 1.$$

These critical exponents helps us to understand the behavior of the system near the critical point and to classify the universality class of the phase transition. Usually when quantities diverge at the critical point, we can characterize their behavior by a power law with a critical exponent, which provides insights into the nature of the fluctuations and correlations in the system as it undergoes a phase transition. Indeed in an experiments, we can measure also **finite size effect**: all of our computations are true in the thermodynamic limit $N \to \infty$, but in a real experiment we have a finite number of particles, so we can observe how the critical behavior is affected by the finite size of the system, which can lead to rounding and shifting of the critical point, as well as modifications to the scaling behavior of physical quantities near the transition. All of these is heavily connected to the concept of **correlation length**, which is a measure of how far correlations between particles extend in the system, and it diverges at the critical point, leading to long-range correlations and critical fluctuations that dominate the behavior of the system near the transition.

How can we concile the mean field predictions for the critical exponents with experimental results? The mean field theory provides is just an approximation, and it is known to give incorrect predictions for the critical exponents in low-dimensional systems (e.g., 2D and 3D), where fluctuations play a significant role (speaking about the Ising model). In higher dimensions (e.g., 4D and above), the mean field theory becomes more accurate, and the critical exponents predicted by mean field theory are closer to experimental results. For Ising

- In 1D, the Ising model does not exhibit a phase transition at finite temperature, and the critical exponents are not defined in the usual sense. The mean field theory incorrectly predicts a phase transition in 1D, which is a clear indication of its limitations in low-dimensionality.
- In 2D, the exact solution of the Ising model gives $\beta = 1/8$ and $\gamma = 7/4$, which are different from the mean field predictions.
- In 3D, numerical simulations and experiments suggest that $\beta \approx 0.326$ and $\gamma \approx 1.237$, which again differ from the mean field values.
- In 4D and above, the critical exponents approach the mean field values of $\beta = 1/2$ and $\gamma = 1$.

Here we can see that the mean field theory provides a qualitative understanding of the phase transition and critical behavior, but it fails to capture the correct critical exponents in low-dimensional systems due to the neglect of fluctuations, which are crucial in determining the behavior near the critical point. More sophisticated approaches, such as renormalization group theory, are needed to accurately describe critical phenomena and compute critical exponents in low-dimensional systems.

We will solve the Ising model in 1D in the next section, and we will give some arguments about the behavior of the model in 2D, which will allow us to understand better the limitations of mean field theory and the importance of fluctuations in low-dimensional systems.

2.2.5 | 1D-Ising Model

The hamiltonian of the 1D Ising model is given by

$$H = -J\sum_{i=1}^{N} s_i s_{i+1} - h\sum_{i=1}^{N} s_i,$$

where we have periodic boundary conditions $s_{N+1} = s_1$. The partition function depends upon the temperature T, the external magnetic field h and the number of spins N, and it can be written as

$$Z_N(\beta, h) = \sum_{\{s_i\}} e^{-\beta H(\{s_i\})}.$$

It is important since we can derive an expression for the ensemble average of the magnetization per site as a function of the partition function:

$$\mathbb{E}_c\left[\frac{1}{N}\sum_{i=1}^{N} s_i\right] = -\frac{1}{N\beta}\frac{\partial \log Z_N}{\partial h}.$$

If we manipulate the expression of the hamiltonian, to express two symmetric sums, we get

$$H = -J\sum_{i=1}^{N} s_i s_{i+1} - \frac{h}{2}\sum_{i=1}^{N} s_i s_{i+1}$$

$$Z = \sum_{\{s_i\}}\prod_{i=1}^{N}(e^{\beta J s_i s_{i+1} + \frac{\beta h}{2} s_i s_{i+1}})\ldots$$

Now we can introduce the **transfer matrix** T, which is a 2×2 matrix that encodes the interactions between neighboring spins and the effect of the external magnetic field. The transfer matrix is defined as

$$T_{s_i, s_{i+1}} = e^{\beta J s_i s_{i+1} + \frac{\beta h}{2}(s_i + s_{i+1})},$$

where s_i and s_{i+1} can take values of ± 1. The partition function can then be expressed as a product of transfer matrices:

$$Z = \sum_{\{s_i\}} T_{s_1,s_2} T_{s_2,s_3} \cdots T_{s_N,s_1} = \mathrm{Tr}(T^N),$$

where the trace is taken over the possible spin states. The transfer matrix method allows us to compute the partition function and other thermodynamic quantities by diagonalizing the transfer matrix and finding its eigenvalues. The largest eigenvalue of the transfer matrix dominates the behavior of the partition function in the thermodynamic limit, and it can be used to derive expressions for the free energy, magnetization, and other physical quantities of interest. This method is particularly powerful for solving one-dimensional models like the 1D Ising model, where it provides an exact solution for the partition function and allows us to analyze the behavior of the system at different temperatures and external fields.

Diagonalization and Phase Transitions

Expressing the partition function in terms of the transfer matrix, we can notice how the computational complexity of the problem is reduced from summing over 2^N configurations to diagonalizing a 2×2 matrix multiplied N times with itself, which is a much more tractable problem. The eigenvalues of the transfer matrix can be computed explicitly, and they determine the thermodynamic properties of the system. We can write the transfer matrix explicitly as

$$T = \begin{pmatrix} e^{\beta J+\beta h} & e^{-\beta J} \\ e^{-\beta J} & e^{\beta J-\beta h} \end{pmatrix},$$

with the idea of diagonalizing this matrix to find its eigenvalues λ_1 and λ_2. Let us note that $T = T^\dagger$, thus we know there exist an orthogonal matrix $O = O^T$ such that $T = ODO^T$, where D is a diagonal matrix containing the eigenvalues of T. The diagonal maatrix D can be expressed as

$$D = \begin{pmatrix} \lambda_+ & 0 \\ 0 & \lambda_- \end{pmatrix},$$

and we can compute the the Nth power of the transfer matrix as

$$T^N = (ODO^T)^N = (ODO^T)(ODO^T)\cdots(ODO^T) = OD^NO^T,$$

where we have used the fact that $O^TO = I$ to simplify the expression. The partition function can then be expressed as

$$Z = \mathrm{Tr}(T^N) = \mathrm{Tr}(OD^NO^T) = \mathrm{Tr}(D^N) = \lambda_+^N + \lambda_-^N,$$

where λ_+ and λ_- are the eigenvalues of the transfer matrix T and we have used the cyclic property of the trace.

Now we can also compute the determinant of the transfer matrix, which is given by

$$\det(T) = \lambda_+\lambda_- = e^{2\beta J} - e^{-2\beta J} = 2\sinh(2\beta J),$$

while for the trace we had

$$\mathrm{Tr}(T) = \lambda_+ + \lambda_- = e^{\beta J+\beta h} + e^{\beta J-\beta h} + 2e^{-\beta J} = 2e^{\beta J}\cosh(\beta h) + 2e^{-\beta J}.$$

The eigenvalues of the transfer matrix can be computed using the characteristic polynomial, which is given by

$$x^2 - \mathrm{Tr}(T)x + \det(T) = 0,$$

which, inserting the expressions for the trace and determinant, can be rewritten as

$$\ldots$$

and in the end we get the eigenvalues as

$$\lambda_\pm = e^{\beta J}\left(\cosh(\beta h) \pm \sqrt{\sinh^2(\beta h) + e^{-4\beta J}}\right),$$

which gives

$$|\frac{\lambda_-}{\lambda_+}| < 1, \quad |\frac{\lambda_-}{\lambda_+}|^N \ll 1.$$

thus we can write the partition function in the thermodynamic limit as

$$Z = \lambda_+^N + \lambda_-^N = \lambda_+^N \left(1 + \left(\frac{\lambda_-}{\lambda_+}\right)^N\right) \approx \lambda_+^N,$$

which allows us to compute the free energy per site as

$$f = -\frac{1}{\beta} \lim_{N\to\infty} \frac{1}{N} \log Z,$$

where $f = \lim_{N\to\infty} \frac{F}{N}$; thus we have

$$f = -\frac{1}{\beta} \log \lambda_+ = -J - \frac{1}{\beta} \log \left(\cosh(\beta h) + \sqrt{\sinh^2(\beta h) + e^{-4\beta J}}\right).$$

from this we can understand that there are no singularities in the free energy as a function of temperature T and external field h, which indicates that there is no phase transition in the 1D Ising model at finite temperature. This is consistent with the known result that the 1D Ising model does not exhibit a phase transition at finite temperature, and it highlights the importance of dimensionality and fluctuations in determining the behavior of the system near critical points.

Let us now compute the magnetization per site as a function of temperature T and external field h. We can use the expression for the partition function to compute the magnetization as

$$\mathbb{E}_c\left[s_i\right] = m = -\frac{1}{N}\frac{\partial}{\partial h} \log Z = -\frac{1}{\beta}\frac{\partial}{\partial h}(\frac{F}{N}).$$

Inserting the expression for the free energy per site, we get

$$m = \frac{\sinh(\beta h)}{\sqrt{\sinh^2(\beta h) + e^{-4\beta J}}}.$$

This expression shows that the magnetization m is a smooth function of temperature T and external field h, and it does not exhibit any singularities or discontinuities, which is consistent with the absence of a phase transition in the 1D Ising model at finite temperature.

TODO: add plot of m vs h for different T.

As T increases, the curve becomes more steep, but it never exihibit a discontinuity or a singularity, which confirms that there is no phase transition in the 1D Ising model at finite temperature. Only in the limit $T \to 0$ we have a discontinuity in the magnetization at $h = 0$, where the system transitions from a fully magnetized state with $m = +1$ for $h > 0$ to a fully magnetized state with $m = -1$ for $h < 0$:

$$\lim_{T\to 0} m(T, h) = \operatorname{sgn}(\sinh(\beta h)) = \operatorname{sgn}(h).$$

We could repeat the computation for χ the magnetic susceptibility, which is given by

$$\chi = \frac{\partial}{\partial h} m(T, h) = \dots$$

and it can be shown that χ does not diverge at any finite temperature, which is consistent with the absence of a phase transition in the 1D Ising model at finite temperature. Only in the limit $T \to 0$ we have a divergence in the magnetic susceptibility at $h = 0$, which indicates a critical point at zero temperature.

The euristic derivation of mean field theory has to have some assumptions which do not hold at low dimensions, and this is the reason why it fails to capture the correct critical behavior in 1D and 2D systems. Only in higher dimensions, where fluctuations are less important, the mean field theory provides a more accurate description of the critical behavior and the critical exponents.

The exact solution of the 1D Ising model using the transfer matrix method allows us to understand the limitations of mean field theory and the importance of fluctuations in determining the behavior of the system near critical points.

Two Point Correlation Function

It is now time to compute the two-point correlation function for the 1D Ising model: the correlation function is defined as

$$\mathbb{E}_c\left[s_i s_{i+r}\right] = \frac{1}{Z}\sum_{\{s_k\}} s_i s_{i+r} e^{-\beta H(\{s_k\})},$$

which can become a connected correlation function if we subtract the product of the magnetizations:

$$\mathbb{E}_c\left[s_i s_{i+r}\right] - \mathbb{E}_c\left[s_i\right]\mathbb{E}_c\left[s_{i+r}\right] = \frac{1}{Z}\sum_{\{s_k\}} s_i s_{i+r} e^{-\beta H(\{s_k\})} - m^2.$$

To compute the correlation function, we can use the transfer matrix method again. We can express the correlation function in terms of the transfer matrix as follows:

$$\mathbb{E}_c\left[s_i s_{i+r}\right] = \frac{1}{Z}\sum_{\alpha_1=1}^{2}\sum_{\alpha_2=1}^{2}\cdots\sum_{\alpha_N=1}^{2} \sigma_{\alpha_j}\sigma_{\alpha_{j+r}} T_{\alpha_1,\alpha_2}T_{\alpha_2,\alpha_3}\cdots T_{\alpha_N,\alpha_1},$$

where σ_{α_j} is the spin value corresponding to the state α_j (i.e., $\sigma_1 = +1$ and $\sigma_2 = -1$). We can rewrite this expression in a more compact form using matrix notation. Let us define a diagonal matrix D such that

$$D = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \text{ and } D_{\alpha,\beta} = \delta_{\alpha,\beta}\cdot\sigma_\alpha,$$

meaning that D is a diagonal matrix with entries corresponding to the spin values. Then we can express the correlation function as

$$\mathbb{E}_c\left[s_j s_{j+r}\right] = \frac{1}{Z}\sum_{\alpha_1=1}^{2}\sum_{\alpha_2=1}^{2}\cdots\sum_{\alpha_N=1}^{2} T_{\alpha_1,\alpha_2}T_{\alpha_2,\alpha_3}\cdots T_{\alpha_{j-1},\alpha_j}D_{\alpha_j,\beta_1}T_{\beta_1,\alpha_{j+1}}\cdots T_{\alpha_{j+r-1},\alpha_{j+r}}D_{\alpha_{j+r},\beta_2}T_{\beta_2,\alpha_{j+r+1}}\cdots T_{\alpha_N,\alpha_1},$$

where we have inserted the diagonal matrices D at the positions corresponding to the spins s_j and s_{j+r}. This expression can be further simplified by recognizing that the sums over the indices α_k correspond to matrix multiplications, and we can express

$$\mathbb{E}_c\left[s_j s_{j+r}\right] = \frac{1}{Z}\mathrm{Tr}(T^{j-1}DT^rDT^{N-r+1-j}),$$

where we have used the cyclic property of the trace to rearrange the order of the matrices. This expression allows us to compute the two-point correlation function in terms of the transfer matrix and the diagonal matrix D, which encodes the spin values. By diagonalizing the transfer matrix with a unitary matrix U:

$$T = U\begin{pmatrix} \lambda_+ & 0 \\ 0 & \lambda_- \end{pmatrix}U^\dagger,$$

where U can be computed explicitly, we can express the correlation function in terms of the eigenvalues and eigenvectors of the transfer matrix, which allows us to analyze its behavior as a function of the distance r and the temperature T. Let us focus on the case without external field: the unitary transformation becomes

$$\cdots$$

and so we can write the transfer matrix in the diagonal basis as

$$\ldots$$

Now putting everything together, we can express the correlation function as

$$\ldots$$

In the end we find a close analytical form for the correlation function, which can be expressed as

$$\mathbb{E}_c\left[s_j s_{j+r}\right]_{h=0} = \frac{\lambda_+^{N-r}\lambda_-^r + \lambda_-^{N-r}\lambda_+^r}{\lambda_+^N + \lambda_-^N},$$

and again if we use tha fact that $|\frac{\lambda_-}{\lambda_+}| < 1$, we can simplify this expression in the thermodynamic limit as

$$\mathbb{E}_c\left[s_j s_{j+r}\right]_{h=0} = \cdots \sim e^{-\frac{r}{\xi}},$$

effectively defining the correlation length ξ as

$$\xi^{-1} = \log\left(\frac{\lambda_+}{\lambda_-}\right) > 0,$$

which is a measure of how far correlations between spins extend in the system. Now considering the absence of an external field, we can express the eigenvalues of the transfer matrix as ...

and in the end we get the correlation length in the $T \to 0$ limit without external field as

$$\xi \sim e^{2\beta J},$$

so that it diveres exponentially as $T \to 0$, which indicates that the system exhibits long-range correlations at low temperatures, and it confirms the presence of a critical point at zero temperature in the 1D Ising model. This behavior of the correlation length is consistent with the absence of a phase transition at finite temperature, as the correlation length remains finite for all $T > 0$ and only diverges as $T \to 0$.

The correlation lenght correspond to the distance after which we can consider the spins as independent, since the correlation function decays exponentially with distance r as $e^{-r/\xi}$. As the temperature approaches zero, the correlation length diverges, indicating that the spins become correlated over longer distances, which is a signature of critical behavior at zero temperature. This analysis of the two-point correlation function and the correlation length provides insights into the nature of correlations in the 1D Ising model and helps to understand the absence of a phase transition at finite temperature. But since this divergence occur at $T = 0$ then everything is in accord to the fact that there is no phase transition at finite temperature, and the critical point is located at zero temperature.

2.2.6 | 2D-Ising Model

We will not solve exactly the 2D Ising model, but we will give some arguments about its behavior and the nature of the phase transition it exhibits, and then in the end we will present the proof of the existence of a phase transition in the 2D Ising model at finite temperature, which is a non-trivial result that highlights the importance of dimensionality and fluctuations in determining the behavior of the system near critical points.

Phase Transition - Euristic

Consider the 2D Ising model on a square lattice, where each spin interacts with its four nearest neighbors. Take the limit for a very low temperature $T \to 0$, and suppose that there is a region in the first 1D raw where the system is in a fully magnetized state, with all l spins aligned (e.g., $m = +1$). We want to ask ourselves if this state is stable against the formation of a droplet of flipped spins (e.g., $m = -1$) of size $l \times l$. The energy cost of creating such a droplet can be estimated as follows:

- Case A: $\uparrow\uparrow \dots \uparrow$, where we have a single domain wall of length l separating the two regions of opposite magnetization. The energy cost of creating this domain wall is proportional to its length, which gives
$$E = -Jl,$$
where J is the coupling constant. The energy cost is negative, which means that it is energetically favorable to create such a droplet, and the system will tend to form domains of flipped spins, leading to a disordered state with zero magnetization.
- Case B: $\uparrow \dots \uparrow\downarrow \dots \downarrow$, where we have two domain walls of length l separating the three regions of magnetization. The energy cost of creating these two domain walls is proportional to their total length, which gives
$$E = -Jl + 2Jl = Jl,$$
where the first term corresponds to the energy cost of creating the first domain wall, and the second term corresponds to the energy cost of creating the second domain wall. The energy cost is positive, which means that it is energetically unfavorable to create such a droplet.

Remark (entropy of domains). ...

If we now compute the free energy cost of creating such a droplet, we need to take into account both the energy cost and the entropy gain associated with the formation of the droplet. The free energy cost can be expressed as

$$F_B - F_A = \Delta E - T\Delta S = 2Jl - Tk_B \log(l-1),$$

since the energy cost of creating the droplet in case B is $2Jl$ and the entropy gain is proportional to the logarithm of the number of ways to arrange the droplet, which is approximately $\log(l-1)$. As l increases, the energy cost dominates over the entropy gain, and the free energy cost becomes positive, which means that it is energetically unfavorable to create such a droplet. This indicates that the system will tend to remain in a magnetized state with non-zero magnetization, and it suggests that there is a phase transition at a finite temperature T_c below which the system exhibits spontaneous magnetization.

This argument provides an intuitive understanding of why the 2D Ising model exhibits a phase transition at finite temperature, and it highlights the importance of dimensionality and fluctuations in determining the behavior of the system near critical points.

It seems that breaking a domain wall is energetically unfavorable, but there is a maximum value of l for which the entropy does not make unfavorable the creation of the droplet, and this maximum value of l can be estimated by setting the free energy cost to zero:

$$\Delta F = 0 \implies 2Jl - Tk_B \log(l-1) = 0 \implies l \sim e^{\frac{2J}{Tk_B}} = \xi,$$

which indicates that there is a characteristic length scale l above which the domain wall are not favorable anymore, and this length scale can be identified with the correlation length ξ of the system. As the temperature approaches the critical temperature T_c from below, the correlation length diverges, indicating that the system exhibits long-range correlations and critical behavior at the phase transition.

Euristic arguments in statistical physics can provide valuable insights into the behavior of complex systems and help to understand the underlying mechanisms driving phase transitions and critical phenomena. Sometimes, when a problem is mathematically intractable, we can use physical intuition and qualitative reasoning to make predictions about the behavior of the system, which can then be used to address the problem in new ways and in the end find the analutical solution.

Peierls's argument - Euristic

TODO: add picture of the two configurations

Let us compare the free energy of the two configurations:

$$\dots$$

We ask what is the difference in energy of the two 2D configurations described. Each set of nearest neighbours of the subdomain considered in the configuration contributes to the change in energy from one configuration to the other. In particular

$$\Delta U = U_B - U_A = 2Jl,$$

where now in 2D l represents the lenght of the perimeter of the flipped subdomain, which is proportional to the number of nearest neighbor pairs that are affected by the change in configuration. The energy cost of creating a droplet of flipped spins is proportional to the length of the domain wall, which is $2Jl$ in this case. This energy cost is positive, which means that it is energetically unfavorable to create such a droplet, and it suggests that the system will tend to remain in a magnetized state with non-zero magnetization, indicating the presence of a phase transition at finite temperature.

But we have always to take into account the entropy gain associated with the formation of the droplet, which can be estimated after counting the number of ways to arrange the droplet of flipped spins. The number of ways to arrange a droplet of size l can be approximated as the path we can take to create the domain wall, which is given by three choices of direction at each step (since we cannot go back),[4] thus we can estimate the number of configurations as

$$\#DM \sim 3^l,$$

thus if the number of domains wall of length l is given by $\#DM$, then the entropy gain associated with the formation of the droplet can be estimated as

$$\Delta S = k_B \log(\#DM) = k_B l \log(3),$$

which is proportional to the length of the domain wall.

Thus in the end, the free energy cost of creating such a droplet can then be expressed as

$$\Delta F = \Delta U - T\Delta S = 2Jl - Tk_B l \log(3) = l(2J - Tk_B \log(3)),$$

which indicates that there is a critical temperature T_c at finite temperature, at which we observe a phase transition.

[4] Extimating like this is exagerated, since we are counting also "snakes" that do not close on a region, but let us do this anyay and then reason on it.

Rigorous Proof of Phase Transition

Let us consider a system with **fixed open boundary conditions**, where we have an Ising model defined on a square lattice of size $N \times N$, and we fix the spins on the boundary to be all up (i.e., $s_i = +1$ for all spins on the boundary). We want to show that there is a phase transition at finite temperature, which can be done by comparing the free energy of two configurations: one with all spins up (the ordered phase) and one with a droplet of flipped spins (the disordered phase).

TODO: Add reference to 14.3 Huang's book stat mech.

The hamiltonian of the system is unchanged, and it is given by

$$H = -J \sum_{\langle i,j \rangle} s_i s_j - h \sum_i s_i,$$

where the first sum is taken over nearest neighbor pairs of spins, and the second sum is taken over all spins in the lattice.

We can fix the temperature and compute the partition function and magnetization for both configurations. Then in the end we will find: if we denote by N_- the average number of down spins, we will have only two cases:

- $\lim_{N\to\infty} \frac{N_-}{N} = \frac{1}{2}$: in this case the system is in a disordered phase with zero magnetization, since the number of up spins and down spins are equal in the thermodynamic limit.
- $\lim_{N\to\infty} \frac{N_-}{N} < \frac{1}{2}$: in this case the system is in an ordered phase with non-zero magnetization, since the number of up spins is greater than the number of down spins in the thermodynamic limit.

We used a preferred direction setting the boundary conditions, but we could have done the same analysis with all spins down at the boundary, and we would have found the same result. Exact numbers will fluctuate, but these ratios will converge to the same limit, and in the TD limit we will have a well defined phase transition at finite temperature.

2.3 | Landau-Ginzburg Effective Field Theory

Appendices

A | Notation and Conventions

B | Computations and further developments